



Task 1 Strategic PV Analysis and Outreach

SEVERE

TRENDS IN PHOTOVOLTAIC APPLICATIONS 2024

REPORT IEA PVPS T1-43:2024

PHOTOVOLTAIC POWER SYSTEMS TECHNOLOGY COLLABORATION PROGRAMME

WHAT IS IEA PVPS TCP?

The International Energy Agency (IEA), founded in 1974, is an autonomous body within the framework of the Organization for Economic Cooperation and Development (OECD). The Technology Collaboration Programme (TCP) was created with a belief that the future of energy security and sustainability starts with global collaboration. The programme is made up of thousands of experts across government, academia, and industry dedicated to advancing common research and the application of specific energy technologies.

The IEA Photovoltaic Power Systems Programme (IEA PVPS) is one of the TCP's within the IEA and was established in 1993. The mission of the programme is to "enhance the international collaborative efforts which facilitate the role of photovoltaic solar energy as a cornerstone in the transition to sustainable energy systems." In order to achieve this, the Programme's participants have undertaken a variety of joint research projects in PV power systems applications. The overall programme is headed by an Executive Committee, comprised of one delegate from each country or organisation member, which designates distinct

'Tasks,' that may be research projects or activity areas. This report has been prepared under Task 1, which deals with market and industry analysis, strategic research and facilitates the exchange and dissemination of information arising from the overall IEA PVPS Programme.

The IEA PVPS participating countries in 2023 are **Australia, Austria, Belgium, Canada, China, Denmark, Finland, France, Germany, Israel, Italy, Japan, South Korea, Malaysia, Morocco, the Netherlands, Norway, Portugal, South Africa, Spain, Sweden, Switzerland, Thailand, Türkiye, and the United States of America.** The European Commission, Solar Power Europe, the Smart Electric Power Alliance (SEPA), the Solar Energy Industries Association, the Solar Energy Research Institute of **Singapore** and Enercity SA are also members.

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AUTHORS

Main Authors: Gaëtan Masson, (Becquerel Institute), Melodie de l'Épine (Becquerel Institute France) Izumi Kaizuka (RTS Corporation)

Analysis: Izumi Kaizuka (RTS Corporation), Elina Bosch, Philippe Macé, Gaëtan Masson, Adrien Van Rechem (Becquerel Institute), Caroline Plaza, Anna Barguès (Becquerel Institute France), Johan Lindahl, Amelia Oller Westerberg (Becquerel Sweden).

Data: IEA PVPS Reporting Countries, Becquerel Institute (BE), RTS Corporation (JP). For the non-IEA PVPS countries Becquerel Sweden.

Editor: Gaëtan Masson, IEA PVPS Task 1 Task Manager.

Design: Mahamadou Tchiffa

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Front cover: Over Easy Solar - Vertical Solar Panels for Green Roofs and Flat Roofs, credit Over Easy Solar

ISBN 978-3-907281-68-0 : Trends in Photovoltaic Applications 2024.



REPORT SCOPE AND OBJECTIVES

The Trends report's objective is to present and interpret developments in the PV power systems market and the evolving applications for these products within this market. These trends are analysed in the context of the business, policy and nontechnical environment in the reporting countries.

This report is prepared to assist those who are responsible for developing the strategies of businesses and public authorities, and to support the development of medium-term plans for electricity utilities and other providers of energy services. It also provides guidance to government officials responsible for setting energy policy and preparing national energy plans. The scope of the report is limited to PV applications with a rated power of 40 W or more. National data supplied are as accurate as possible at the time of publication. Data accuracy on production levels and

system prices varies, depending on the willingness of the relevant national PV industry to provide data. This report presents the results of the 27th international survey. It provides an overview of PV power systems applications, markets and production in the reporting countries and elsewhere at the end of 2023 and analyses trends in the implementation of PV power systems between 1992 and 2023. Key data for this publication were drawn mostly from national survey reports and information summaries, which were supplied by representatives from each of the reporting countries. Information from the countries outside IEA PVPS are drawn from a variety of sources and, while every attempt is made to ensure their accuracy, the validity of some of these data cannot be assured with the same level of confidence as for IEA PVPS member countries.

ACKNOWLEDGMENT

This report has been prepared under the supervision by Task 1 participants. A special thanks to all of them.

FOREWORD

2023 was another record year for PV development and the energy transition but also a tumultuous one whose effects will have long-lasting impact on the PV industry.

Installations reached the astonishing value of 456 GW in a contrasted market: while the Chinese PV market grew significantly to absorb its own industry's overcapacities, the rest of the global PV market experienced a strong, but very much less impressive development to reach 181 GW. The year-on-year change brought the Chinese PV market well above the sum of all other markets globally, concentrating PV development at a never-seen level.

More than 1.6 TW of PV systems were operational at the beginning of the year 2024, producing more than 2 135 TWh of electricity, or 8.3% of the global electricity demand. The contribution of PV to reducing greenhouse gases emissions amounted to around 0.92 GTons of CO₂, or 2.5% of all emissions from the energy sector if we consider it now replaces baseload generation. This astonishing number confirms the status of PV as a key energy source of this century and its ability to provide affordable, decarbonized and scalable electricity suitable for all locations and system sizes.

PV is reaching maturity whilst diversifying its uses: agrivoltaics are one the key features of the year 2023, with developers of PV systems embracing the need to ensure a sustainable dual usage of land, enhancing or maintaining agriculture while adding energy production. In numerous regions, storage already improves the dispatchability of PV electricity whilst supporting stable networks and allowing increasingly high penetration rates of PV. Massive plans have been announced to either transport electricity to distant consumption sites or transform competitive PV electricity into liquid fuels. Green hydrogen and its derivatives aiming at feeding the industry with cleaner hydrogen or other chemicals could, in the coming years, open a path for starting the decarbonization of complex industrial processes.

With 7 million jobs in the solar PV sector, the contribution to the global economy has significantly accelerated in the last years, and PV contributes massively to the economy of numerous countries.

However, as 2023 has shown, the development of PV in countries with a rather stable power mix suffered from setbacks and delays. While China was accelerating its development, other countries like the USA or India increased volumes but could have achieved even higher installation levels, had various administrative, political or local issues, starting with weak grids or limited social acceptance, been resolved. Other countries couldn't match the Chinese acceleration and the PV manufacturing industry suffered huge overcapacities that resulted in a complete collapse of the price of PV components, especially modules. These low prices continue in 2024 and threaten the viability of the entire PV manufacturing industry, not only in China, but in other countries and regions where projects that were intended to create local jobs and involve populations in the development of PV have been at risk.

This temporary imbalance between manufacturing supply and demand has put tremendous pressure on the manufacturing industry and will probably lead consolidation and possible bankruptcies. The current (mid 2024) low prices can be considered as unsustainable, however the competitiveness of PV was already guaranteed on mid 2023 prices, and prospects for a fast development in the coming years remain bright in many countries.

Gaëtan Masson

Manager Task 1
IEA PVPS Programme

Daniel Mugnier

Chair
IEA PVPS Programme



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TRENDS IN PHOTOVOLTAIC APPLICATIONS // 2024

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TOTAL BUSINESS VALUE IN PV SECTOR IN 2023

\$400 BILLION USD



TOP 5

PV MARKETS IN 2023

	CHINA	277 GW
	EU	59.8 GW
	USA	33.9 GW
	INDIA	13.0 GW
	BRAZIL	12.4 GW

PV CONTRIBUTION TO ELECTRICITY DEMAND

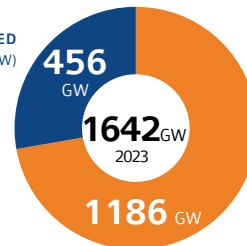


8.3%

Share of PV in the global electricity demand in 2023

ANNUAL INSTALLED CAPACITY IN 2023 (GW)

GLOBAL PV CAPACITY END OF 2023



GLOBAL PV CAPACITY END OF 2022 (GW)

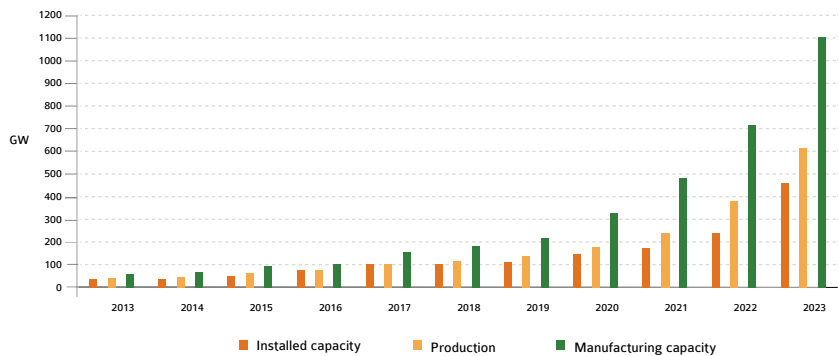
CLIMATE CHANGE IMPACTS



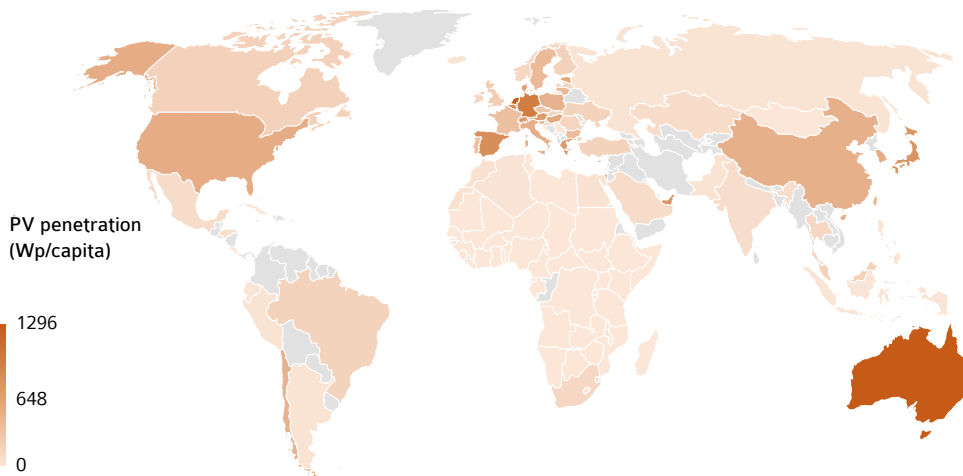
923

million tons of CO₂ saved in 2023
* method changed from 2022; now assuming PV replaces baseload generation

YEARLY PV INSTALLATION, MODULE PV PRODUCTION AND MODULE PRODUCTION CAPACITY 2013-2023 (GW)



PV PENETRATION PER CAPITA IN 2023



54 COUNTRIES REACHED AT LEAST

1 GWp

IN 2023

PV POWER PER CAPITA

1. AUSTRALIA (1 296 Wp/CAP)
2. THE NETHERLANDS (1 288 Wp/CAP)
3. GERMANY (974 Wp/CAP)
4. BELGIUM (842 Wp/CAP)
5. SPAIN (806 Wp/CAP)

36 COUNTRIES INSTALLED AT LEAST

1 GWp

IN 2023

SOURCE IEA PVPS & OTHERS



one

INTRODUCTION TO THE CONCEPTS AND METHODOLOGY

PV TECHNOLOGY

Photovoltaic (PV) devices convert light directly into electricity and should not be confused with other solar technologies such as concentrated solar power (CSP) or solar thermal for heating and cooling. The key components of a PV power system are various types of photovoltaic cells (often called solar cells) interconnected and encapsulated to form a photovoltaic module (the commercial product), the mounting structure for the module or array, the inverter (one or multiple, essential for grid-connected systems and required for most off-grid systems) the cabling to connect the modules with each other and with the inverter(s), the storage battery and charge controller (for off-grid systems but also increasingly for grid-connected ones).

CELLS, MODULES AND SYSTEMS

Photovoltaic cells represent the smallest unit in a photovoltaic power producing device. Wafer sizes, and thus cell sizes, have progressively increased, as this is considered by industrial actors to be an easy way to improve cell and module wattage. Until recently, wafer sizes ranged from 156.75 x 156.75 square mm (named M2) up to 210 x 210 square mm (named M12). In 2023, a variety of cell sizes and shapes were present on the market, although a majority of cells were 210 mm square and 182 mm square. However, in 2023 and 2024, major manufacturers came to a consensus and agreed to standardise towards rectangular cells, either the 182R - a cell with a short side of 182mm and a long side from 188 mm up to 210 mm – or the 210R (182 mm x 210 mm) and consequently these are expected to progressively replace the 182 mm square or 210 mm square present on the market in 2023. In general, cells can be classified as wafer-based crystalline silicon

(c-Si) (mono- and multi-crystalline), compound semiconductor (thin-film), or organic.

Today, c-Si technologies account for more than 98% of the overall cell production. Monocrystalline or Single crystalline PV cells, formed with wafers manufactured using a single-crystal growth method, feature commercial efficiencies between 20% and 25% (single junction). The silicon PV module market is nearly exclusively composed of these cells (approaching 100%), as they have replaced multicrystalline silicon (mc-Si) cells, also called polycrystalline. These were formed with multicrystalline wafers, manufactured by a cast solidification process. They were less efficient, with an average conversion efficiency of approximately 18% - 21% in mass production (single-junction) and although there are nearly no new modules with this technology entering the market, they are present in a large volume of already operational PV systems.

Thin-film cells are formed by depositing extremely thin layers of photovoltaic semiconductor materials onto a backing material such as glass, stainless steel or plastic. III-V compound semiconductor PV cells are formed using materials such as Gallium Arsenide (GaAs) on Germanium (Ge) substrates and have high conversion efficiencies from 25% up to 30% (not concentrated). Due to their high cost, they are typically used in space applications. Thin-film modules used to have lower conversion efficiencies than basic crystalline silicon technologies, but this has changed in recent years. They are potentially less expensive to manufacture than crystalline cells thanks to the reduced number of manufacturing steps from raw materials to modules, and to reduced energy demand.

PV TECHNOLOGY / CONTINUED

Thin-film materials commercially used are cadmium telluride (CdTe), and copperindium-(gallium)-diselenide (CIGS and CIS). Amorphous (a-Si) and micromorph silicon (μ -Si) used to have a significant market share but failed to follow both the price decline of crystalline silicon cells and the efficiency increase of other thin film technologies. The thin-film cell technology efficiency ranges between 14% (OPV)¹, 12% (a-Si), 19.2% (CIGS and CIS), 19.9% (CdTe), 25.1% single junction GaAs and 31.2% three-junction GaAs (non-concentrated) and above 35% for some CPV modules. It should be noted that cell conversion efficiencies are generally 2.5% higher than the commercial module efficiency indicated here.

Organic thin-film PV (OPV) cells use dye or organic semiconductors as the light-harvesting active layer. Organic and inorganic hybrid materials such as perovskites are also used for photovoltaic materials. Perovskite technology has created increasing interest and research over the last few years and is currently the fastest-advancing solar technology. Despite the potentially low production costs, stable products are difficult to manufacture, nevertheless development and demonstration activities are underway. Tandem cells based on perovskites are an important focal point of current research, with either a crystalline silicon base or a thin film base and could hit the market sooner than pure perovskites products. In 2023, an experimental perovskite solar cell achieved 33.9% efficiency in silicon-based tandem and 24.9% efficiencies in CIGS or CIS-based tandems. Several Chinese manufacturers have announced shipping perovskite modules in 2022 and 2023.

Photovoltaic modules are typically rated from 350 W to 600 W, or even up to 740W in 2023 for bifacial glass modules, depending on the technology and the size – although typical sizes for residential systems in 2023 was 350 W to 435 W, with larger modules above 540 W more often reserved for centralised ground mounted systems. Specialized products for building integrated PV systems (BIPV) exist, sometimes with higher nominal power due to their larger sizes. Crystalline silicon modules consist of individual PV cells connected and encapsulated between a transparent front, usually glass, and a backing material, usually plastic or glass. Thin-film modules encapsulate PV cells formed into a single substrate, in a flexible or fixed module, with transparent plastic or glass as the front material.

A PV system consists of one or several PV modules, connected to either an electricity network (grid-connected PV) or to a series of loads (off-grid).

It comprises various electric devices aimed at adapting the electricity output of the module(s) to the standards of the network/ grid or the load: inverters, charge controllers or batteries.

A wide range of mounting structures have been developed — especially for BIPV — including PV facades, sloped and flat roof mountings, integrated (opaque or semi-transparent) glass-glass modules and PV tiles.

Single or two-axis tracking systems are attractive for ground-mounted systems, particularly for PV utilization in countries with a high share of direct irradiation. By using such systems, the energy yield can be increased from 15% to 35% for single axis and 25% to 50% for dual axis trackers compared with fixed systems. The precise gain depends not only on the latitude of the system, but also the orientation of single axis trackers and the eventual controls and algorithms used to manage the tracking system

PV APPLICATIONS AND MARKET SEGMENTS

When considering distributed PV systems on buildings, it is necessary to distinguish building applied photovoltaics (BAPV) and buildings integrated photovoltaics (BIPV) systems. BAPV refers to PV systems installed on an existing building as an addition to the existing envelope while BIPV implies that the PV replaces conventional building materials such as roofing elements or facades. IEA PVPS Task 15 has reviewed several definitions and versions of how to define BIPV, and compiled it into one:

“A BIPV module is a PV module and a construction product together, designed to be a component of the building. A BIPV product is the smallest (electrically and mechanically) non-divisible photovoltaic unit in a BIPV system which retains building-related functionality. If the BIPV product is dismantled, it would have to be replaced by an appropriate construction product.

A BIPV system is a photovoltaic system in which the PV modules satisfy the definition above for BIPV products. It includes the electrical components needed to connect the PV modules to external AC or DC circuits and the mechanical mounting systems needed to integrate the BIPV products into the building.”

And the same definition is used in this report. Amongst BIPV solutions, PV tiles, or PV shingles, are typically small, rectangular solar panels that can be installed alongside conventional tiles or slates using a traditional racking system used for this type of building product. BIPV products can take various shapes, colours

¹ Module efficiencies are reported from the NREL Champion Module Efficiencies for small or standard modules: <https://www.nrel.gov/pv/module-efficiency.html>



and sizes and be manufactured using various materials, although a vast majority use a glass sandwich composition. They generally replace conventional building envelope solutions, or, less often, provide elements of architectural interest.

Bifacial PV modules collect light on both sides of the panel. Depending on the reflection of the ground underneath the modules (albedo), the energy production increase is estimated to be around 15% but may reach a maximum of up to 30-35% with single axis tracking systems. Bifacial modules have a growing competitive advantage despite higher overall installation costs, and it is estimated that more than 90% of cells deployed in 2023 were bifacial, however many were assembled in monofacial modules, leaving approximately 50% of the 2023 annual PV market composed of bifacial modules. Some challenges remain in being able to accurately simulate the performance of bifacial modules.

Floating PV systems are mounted on a structure that floats on a water surface and can be associated with existing grid connections, for instance when in the vicinity of a hydro power dam. The development of floating PV on man-made water areas is a solution to land scarcity problems in high population density areas and presents other advantages including reduced evaporation rates in dry climates and improved cooling of PV modules for better efficiency in warm climates. Off-shore floating PV systems are installed in several places.

Agricultural PV combines crops and energy production on the same site. PV can either be a static tool added into pastures or crops or a dynamic tool to facilitate agricultural production. The sharing of light between these two types of production potentially allows a higher crop yield, depending on the climate and the selection of the crop variety and can even be mutually beneficial in some cases, as the water which evaporates from the crops can contribute to a reduction of the PV modules' operating temperature. When combined with grazing, the shade provided by modules can increase grass quality in some climates whilst livestock grazing reduces maintenance costs for the PV system.

PV thermal hybrid solar installations (PVT) combine a solar module with a solar thermal collector, converting sunlight into electricity and capturing the remaining waste heat from the PV module to produce hot water or feed heating systems. The water circulating in the modules can reduce the operating temperature of the modules, which benefits the global performances of the system.

Vehicle integrated PV (VIPV) designates the integration of solar cells into the shell of vehicles to reduce emissions in the mobility sector. Solar cell technological developments allow new models to meet both aesthetic expectations for car design and technical requirements, such as light weight and resistance to load. Vehicle

applied PV (VAPV) relates to the use of PV modules on vehicles without integration.

Infrastructure Integrated PV (IIPV) is when PV is integrated into infrastructure such as noise barriers, dam walls, pavement, roads, etc. New applications are being demonstrated regularly.

Solar Home Systems (SHS) or pico PV systems combine the use of efficient lights (mostly LEDs) with charge controllers and batteries. With a small PV panel of only a few watts, essential services can be provided, such as lighting, phone charging and powering a radio or a small computer. Expandable versions of pico PV systems have entered the market and enabled starting with a small kit and adding extra loads later. They are mainly used for off-grid basic electrification, predominantly in developing countries.

GRID-CONNECTED PV SYSTEMS

In grid-connected PV systems, an inverter is used to convert electricity from direct current (DC) as produced by the PV array to alternating current (AC) that is then supplied to the electricity network. The typical weighted conversion efficiency is in the range of 95% to 99%. Most inverters incorporate a Maximum Power Point Tracker (MPPT), which continuously adjusts the load impedance to provide the maximum power from the PV array. One inverter can be used for the whole array or separate inverters may be used for each string of modules. PV modules with integrated inverters, usually referred to as "AC modules", can be directly connected to the electricity network (where approved by network operators). They offer better partial shading management and installation flexibility. Similarly, micro-inverters connected to up to four panels also exist, and despite their higher initial cost, they present some advantages where array sizes are small and maximal performance is to be achieved. "AC modules" could see increasing use in residential systems but also linear PV systems where savings can be made on cable costs when using AC modules. In some specific projects, DC to DC inverters are used as the electricity generated is injected to DC lines such as tram and railway networks.

Grid-connected distributed PV systems are installed to provide power to a grid-connected customer or directly to the electricity network - nearly always the distribution network but, for the largest utility scale systems, sometimes the transmission network. Such systems may be on, or integrated into, the customer's premises - often on the demand side of the electricity meter, on residential, commercial or industrial buildings, or simply in the built environment on motorway sound-barriers, etc. Size is not a determining feature - while a 1 MW PV system on a rooftop may be large by PV standards, this is not the case for other forms of distributed generation.

PV APPLICATIONS AND MARKET SEGMENTS / CONTINUED

Grid-connected centralized PV systems (also called utility scale systems) perform the functions of centralized power stations. The power supplied is not physically associated with an electricity customer, and the system is not located to specifically perform functions on the electricity network other than the supply of bulk power. These systems are typically ground-mounted and function independently of any nearby development.

Hybrid systems combine the advantages of PV and diesel generation in mini grids. They allow mitigating fuel price increases, deliver operating cost reductions, and offer higher service quality than traditional single-source generation systems. Increasingly, diesel generators are being reserved for worst case situations as battery storage becomes cheaper. Combining these technologies provides a reliable and cost-effective power source in remote places such as for telecom base stations. Large-scale hybrids can be used for large cities powered today by diesel generators and have been seen, for instance in central Africa, often in combination with battery storage.

OFF-GRID PV SYSTEMS

For most off-grid systems, a storage battery is required to provide energy during low-light periods. Several battery technologies for off-grid PV systems are commonly commercialised in 2023 including different types of lead-acid batteries and lithium-ion. Each type of battery has specific advantages. The lifetime of a battery varies, depending on the operating regime and conditions, but is typically between 5 and 15 years depending on the technology, usage and maintenance. For some specific applications – typically, water pumping – no storage battery is needed, and energy is consumed as it is generated.

A charge controller (or regulator) is used to maintain the battery at the highest possible state of charge and provide the user with the required quantity of electricity, while protecting the battery from deep discharge or overcharging. Some charge controllers also have integrated MPP trackers to maximize the PV electricity generated. If there is a requirement for AC electricity, a “stand-alone inverter” can supply conventional AC appliances.

Off-grid domestic systems provide electricity to households and villages that are not connected to the utility electricity network. They provide electricity for lighting, refrigeration and other low power loads, have been installed worldwide and are increasingly the most competitive technology to meet the energy demands of off-grid communities.

Off-grid non-domestic installations were the first commercial application for terrestrial PV systems. They provide power for a wide range of applications, such as telecommunications, water pumping, vaccine refrigeration and navigational aids. These are applications where small amounts of electricity have a high value, thus making PV commercially cost competitive with other small generating sources.



METHODOLOGY FOR THE MAIN PV MARKET DEVELOPMENT INDICATORS

This report counts all PV installations, both grid-connected and reported off-grid installations. By convention, the numbers reported refer to the nominal power of PV systems installed. These are expressed in W (or Wp) or Wdc. Several methodological steps are taken when compiling data.

Power capacity reported in AC is converted to DC (nominal) power when necessary to calculate the most precise installation numbers every year: Some countries report the power output of the PV inverter or even the power of the grid connection. The difference between the standard DC Power (in Wp) and the AC power can range from as little as 5 % (conversion losses) to as much as 40 % (for instance some grid regulations limit output to as little as 65 % of the peak power from the PV system, but also higher DC/AC ratios reflect the evolution of utility-scale PV systems). For some countries, this means publishing different values to official data – for example, China’s National Energy Administration (NEA) publishes in AC and PVPS applies a conversion ratio from AC to DC. A range of values is often provided to account for uncertainty in AC/DC conversion ratios, in particular with regards to new utility scale capacity in China, where the minimal annual volume considers official China reporting, and the maximal annual volume considers a further 42 GW that could have been installed considering the uncertainty surrounding official conversion ratios from AC to DC of utility scale systems. For many figures, these two values have been represented with full (minimum) and additional shaded (maximum) bars. If no range is specified, compiled data refers to the higher totals for China. (Note that the IEA PVPS Snapshot published in April 2024 used the lower of these two values).

Inclusion of countries in geographical or political blocs and regions (see Annex 4): for tables and graph reporting data for major markets, data from member countries of the European Union is reported. When referencing regions, data from all countries on the European continent is aggregated. Unless specified otherwise, the term Europe includes all countries on the continent. The European Commission is a member of the IEA PVPS however this report no longer counts all EU members in IEA PVPS blocs unless specified. Chinese Taipei refers to the economic region, Türkiye is included in Europe, and Korea refers to South Korea.

Global data should be considered as indications rather than exact statistics. Data from countries outside of the IEA PVPS network have been obtained through different sources, some of them based on custom trade statistics.

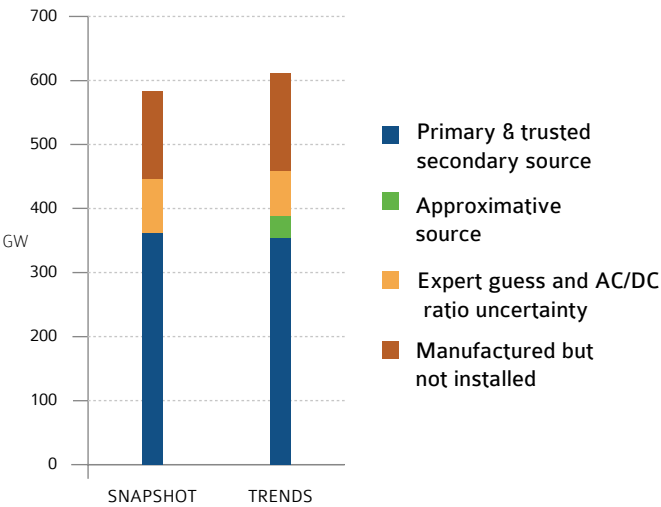
As the PV market grows constantly, reporting of PV installations is becoming more complex. IEA PVPS has decided to count all PV installations, both grid-connected and off-grid, when numbers are reported, and to estimate the remaining part on unreported installations. For countries with historically significant capacity and good reporting, a slow yet growing gap between shipped/imported capacity and installed capacity can be attributed to several factors including conversion factors from AC to DC, repowering and decommissioning. The extremely fast paced development of micro systems (plug&play systems with only a few modules), whilst not significant in overall volumes is symptomatic of the development of unreported systems reaching the market and sometimes being invisible to distribution system operators (DSO) and data collection.

METHODOLOGY FOR THE MAIN PV MARKET DEVELOPMENT INDICATORS / CONTINUED

Other market evolutions such as off-grid applications are difficult to track even in member countries, and significant growth in installations in third countries without a robust reporting system is also a likely source of underreporting. In light of this, reporting here takes into account reported and expert estimates of new commissioned capacity as well as probable unreported volumes installed in one of the above contexts. Data on estimated shipped capacity from custom sources, in inventories, has been incorporated in Figure 1.1 to improve market visibility. As can be seen, between estimations published in the Snapshot in April 2024 and those in this publication, many countries have firmed their evaluation of annual installed volumes. In particular, volumes for India and smaller countries have been revised.

With significant overcapacity in manufacturing, it should be noted that approximately 150 GW of capacity was manufactured but not commissioned; some of this volume has been installed and will be included in 2024 figures as it is commissioned; some of this volume is in inventories in Europe and, to a lesser extent, other countries, whilst the remainder is in export inventories or manufacturers inventories in China.

FIGURE 1.1: ANNUAL INSTALLED VS ANNUAL SHIPPED VOLUMES AND EVOLUTION OF DATA ESTIMATIONS



SOURCE IEA PVPS & OTHERS

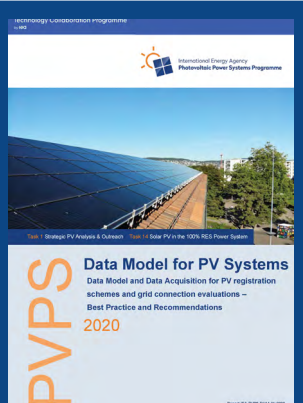
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PV MARKET DEVELOPMENT TRENDS

Over 1.64 TW of PV plants have been installed globally, of which over 50% has been installed in the past three years. In 2023, more than 35 countries had a GW-scale annual market. Whilst the number of national markets with measurable contributions to global PV capacity is increasing every year, the concentration of the market in China over the past three years has decreased their relative importance.

A large majority of PV installations are grid-connected and feed electricity into either the consumer's internal electrical circuit or the electrical grid. PV installation data is reported in DC by default in this report, and this report converts any data officially reported in AC to DC to maintain coherency. When official reporting is in AC, announced capacities may be specified as MWac or MWdc in this report, if necessary, however by default, MW implies MWdc. See Chapter 1 for more information on data conversion methodology, uncertainty and in particular the impact on evaluations of China market figures. For more information on registering PV installations, download the IEA PVPS report on registering PV installation

Download the "Data Model for PV Systems" reports



THE GLOBAL INSTALLED CAPACITY

Global PV installed capacity (GW)
+38% YoY growth

At the end of 2023, the cumulative global installed capacity reached 1 642.0 GW¹ – consistent with the preliminary estimation of 1 624 GW published in the IEA-PVPS Snapshot of Global PV Markets 2023 in April of this year.

It appears that 413.8 GW represented the **minimum** capacity based on trustable sources, and 456.0 GW represented the **most probable** capacity installed during 2023. This volume is almost double that of 2022, itself well above 2021 volumes – resulting from a combination of increased action on climate imperatives, plummeting module costs and actions in China to absorb manufacturing capacity.

1. China's National Energy Administration (NEA) publishes in AC and PVPS applies a conversion ratio from AC to DC. A range of values is often provided to account for uncertainty in AC/DC conversion ratios, in particular with regards to new utility scale capacity in China, where the minimal annual volume considers official China reporting and the maximal annual volume considers a further 42 GW that could have been installed considering the uncertainty surrounding official conversion ratios from AC to DC of Utility scale systems. For many figures, these two values have been represented with full (minimum) and additional shaded (maximum) bars. If no range is specified, compiled data refers to the higher totals with Official China reporting values. Note that the IEA PVPS Snapshot published in April 2024 used the lower of these two values

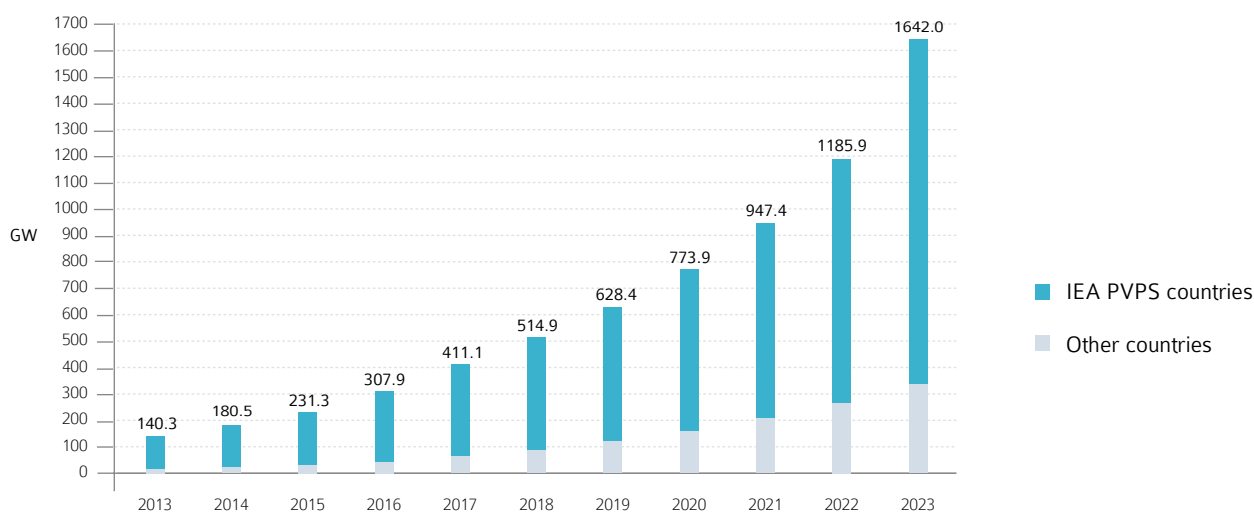
THE GLOBAL INSTALLED CAPACITY / CONTINUED

Among the trustable sources, the group of IEA PVPS² countries represented 1305.7 GW of the global installed capacity. The IEA PVPS participating countries in 2023 are Australia, Austria, Belgium, Canada, China, Denmark, Finland, France, Germany, Israel, Italy, Japan, South Korea, Malaysia, Morocco, the Netherlands, Norway, Portugal, South Africa, Spain, Sweden, Switzerland, Thailand, Türkiye, and the United States of America (USA). The European Commission is a member of the IEA PVPS and, as such, in this report its member countries are counted in IEA PVPS block.

2. For the purpose of this report, IEA PVPS countries are those that are members in their own right – or, as occasionally mentioned explicitly those that are a member in their own right or through the adhesion of the EC.

The other key markets that have been considered and which are not part of the IEA PVPS Programme represented a total cumulative capacity of 336.2 GW at the end of 2023. Amongst them, India still covered around one third of this capacity with 92.6 GW whilst Brazil (37.8 GW), Vietnam (18.6 GW) and the UK (16.0 GW) scored the three first places. These countries represented from 5% (UK) to 28% (India) of the cumulative non-IEA PVPS capacity. Other non-IEA PVPS countries to have significant cumulative volumes include MENA countries with world-record size systems coming progressively online such as the UAE (7.1 GW) and Saudi Arabia (2.9 GW), Chinese Taipei (10.4 GW) and Pakistan (4.5 GW) in Asia. In Europe, Poland (17.4 GW) and Greece (7.2 GW) are the largest cumulative markets, followed by Ukraine (6.2 GW), although here the market has stalled given the ongoing conflict. Another five non-IEA PVPS European countries have cumulative capacity over 2 GW.

FIGURE 2.1: EVOLUTION OF CUMULATIVE PV INSTALLATIONS



SOURCE IEA PVPS & OTHERS

PV PENETRATION PER CAPITA

PV penetration can be measured either as a ratio of Wp per capita or kWh generated to meet a country's electricity demand – here we look at the volume of PV capacity relative to the country's population, indicating the relative efforts made by different countries.

Australia has reached and maintained its place as the highest installed PV capacity per inhabitant with 1 296 W/cap (up 11% on 2022) in IEA-PVPS and surveyed countries. The Netherlands is catching up quickly but is once again second with 1 288 W/cap (+25%). Germany (974 W/cap), Belgium (842 W/cap), Spain (806 W/cap) and Japan (734 W/cap) kept their top spots. Some big

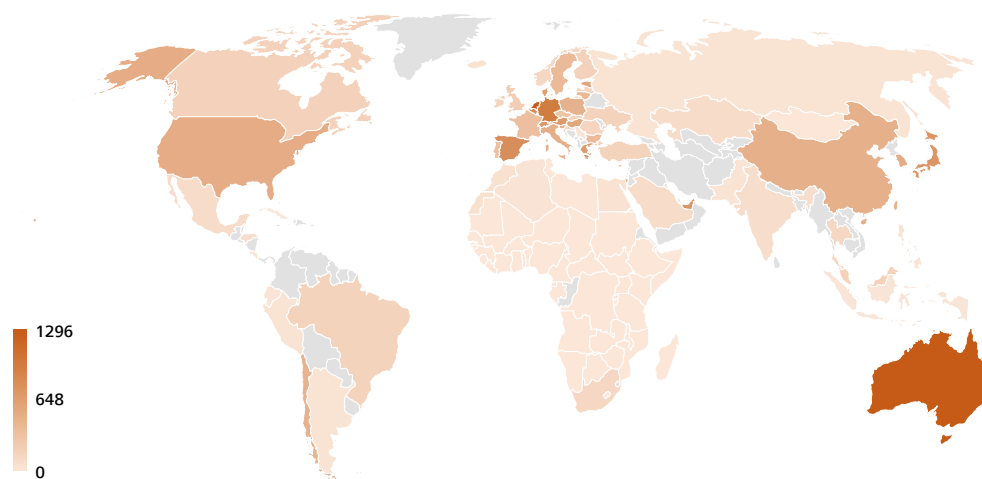
utility-scale systems commissioned in 2023 have allowed the UAE to double their rate to 751 W/cap. Another 10 countries now have more than 500 W/cap, including the USA. With the large volumes installed in China, their penetration rate jumped 50% up to 490 W/cap. Of the largest country markets, India (65 W/cap) and Brazil (175 W/cap) still lag behind.

Typical residential systems have modules with an individual power of 350 Wp to 435 Wp – and now nearly 30 countries have the equivalent of 1 to 3 modules installed per person.

Both Australia and the Netherlands have installed more than 1 250 W/cap.



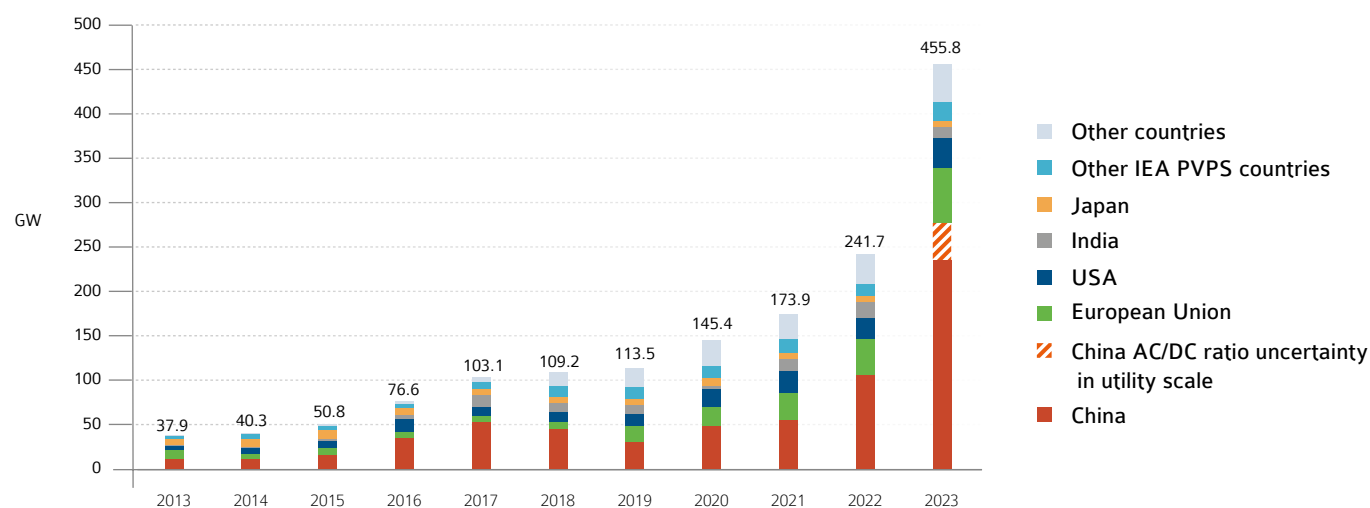
FIGURE 2.2: PV PENETRATION PER CAPITA IN 2023



SOURCE IEA PVPS & OTHERS

EVOLUTION OF PV ANNUAL INSTALLATIONS

FIGURE 2.3: EVOLUTION OF ANNUAL PV INSTALLATIONS IN MAJOR MARKETS



SOURCE IEA PVPS & OTHERS

THE GLOBAL INSTALLED CAPACITY / CONTINUED

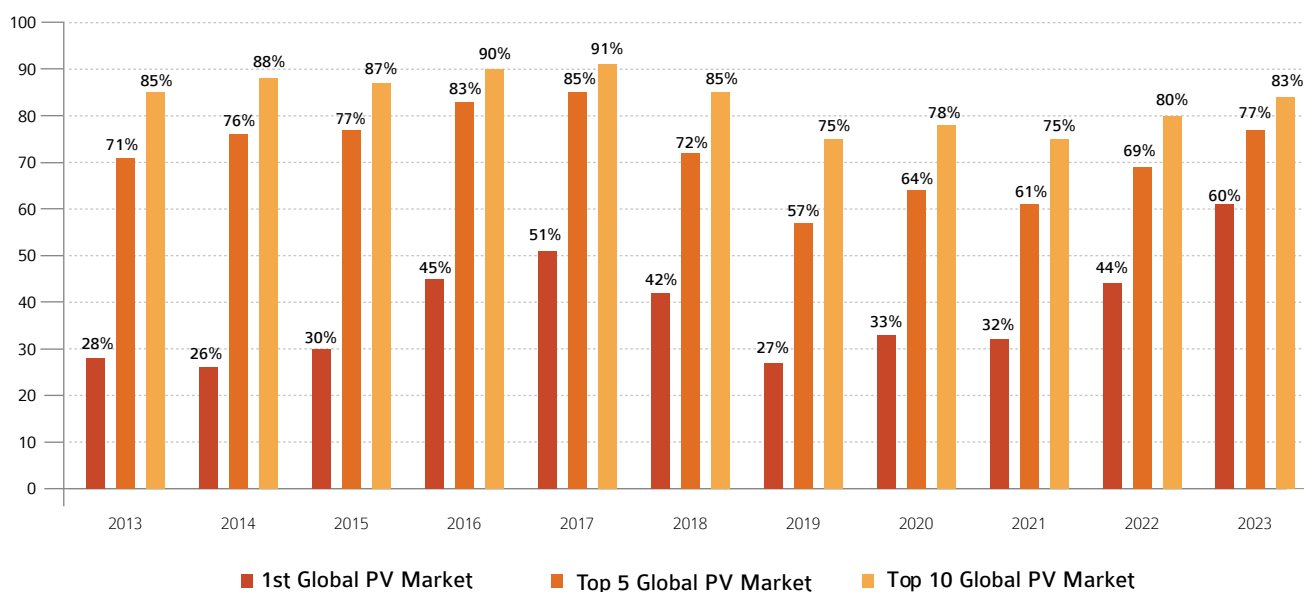
The IEA PVPS countries installed at least 385.8 GW in 2023. While they are more difficult to track with a high level of certainty, installations in non-IEA PVPS countries contributed an estimated amount of 70.2 GW. After two years of what was perceived as strong growth, 2023 beat all the records, nearly doubling 2022 annual capacity as module prices dropped. Consumer, and investor confidence in the ability of PV to provide reliable and stable electricity generation costs as a shelter from electricity market fluctuations was maintained.

Much of the growth has been achieved in China, where installed volumes jumped through the year, although they struggled to keep up with the volume of manufacturing capacity available. Between 235.0 GW and 277.2 GW was installed (converted from China's

National Energy Administration's AC figures and depending on the AC/DC conversion ratio used). Whilst there is an increasingly large uncertainty around the nominal capacity installed in utility-scale plants (the inverter/module ratio is estimated based on industry practices and a few percentage points error can lead to tens of GW of difference in estimated volumes), it is clear that the Chinese market was driven by the centralised segment this year, with just 35% of new capacity in the distributed segment. The Chinese market represented more than 60% of the global installations in 2023, up from last year's 45%. This market dominance is similar to the role Germany played through the mid-late 2000's, and the long-term impacts are difficult to judge. The cumulative capacity installed in China reached at least 649.0 GW and as much as 691.2 GW.

Taken as a bloc, the **European Union** would come in second after China in terms of annual installed capacity, maintaining strong growth rates for the fifth year in a row. The combined annual new capacity was 58.2 GW, led by **Germany** that doubled its market to reach 15.0 GW, **Spain** with a steady 8.9 GW and strong contributions from **Italy** (5.3 GW), **Poland** (4.9 GW) and the **Netherlands** (4.8 GW).

FIGURE 2.4: EVOLUTION OF MARKET SHARE OF TOP COUNTRIES



SOURCE IEA PVPS & OTHERS



The USA annual market grew nearly 50% reaching 33.9 GW, after a slower 2022 (23.1 GW) and 2021 (24.8 GW). Some of the issues that hindered the 2022 market seem to have been partially resolved, including installation backlogs due to supply chain issues. The market remains a utility-scale market, with 24.3 GW or just over 70% of new capacity centralised. The decentralised market saw a robust 9.5 GW installed, solidly above 2022's 8.2 GW. By the end of 2022, the USA reached 177.3 GW of cumulative installed capacity.

After revisions to data for both Germany and India since the publication of the IEA PVPS Snapshot in April, Germany has made it back into the top 3 countries for the first time since 2011, after doubling 2022 volumes to install 15.0 GW, for a total cumulative capacity of 82.3 GW. India was in fourth place with 13.0 GW installed, up from the previous year's 18.1 GW, for a total cumulative capacity of 92.6 GW.

Whilst the market in Brazil didn't maintain the high growth rates of the previous four years, it was still a growth market with 12.4 GW, reaching 37.8 GW cumulative capacity, confirming its place as a globally important market.

With the dominance of the Chinese market, it is no surprise that the market share of the top countries is becoming more and more

elevated - smaller markets are contributing proportionally less to global installation numbers than the major markets. China has done much to absorb the results of its manufacturing capacity growth.

Behind the top 5, Spain in 6th place, (8.9 GW for 38.90 GW cumulative capacity) and Japan in 7th place (6.3 GW for 91.4 GW cumulative capacity) remained major (steady) markets. Italy made it back to the top 10 on the back of strong growth, with 5.3 GW new capacity for a cumulative capacity of 30.3 GW. Both Poland and the Netherlands had similar volumes in 2023 to 2022: Poland with 4.9 GW annual for 17.4 GW cumulative capacity and the Netherlands 4.8 GW of new installations for a total of 23.0 GW.

Together, these 10 markets cover around 84% of the 2023 annual world market, a sign that the growth of the global PV market has been driven by a limited number of countries once again. Market concentration has been fuelling fears for the market's stability in the past, if one of the top three or top five markets would experience a slowdown, although the past years have shown that when one market slows, another is often in growth (witness USA/India, for example). As shown in Figure 2.4, the market concentration steadily decreased in 2019 before growing again in 2020, stabilising in 2021, then growing even more in 2022 and 2023, due to the growth of the Chinese PV market. As new markets are starting to emerge, the concentration of the global PV market minus China

TABLE 2.1: EVOLUTION OF TOP 10 MARKETS

RANKING	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
1	CHINA	CHINA	CHINA	CHINA	CHINA	CHINA	CHINA	CHINA	CHINA	CHINA	CHINA
2	JAPAN	JAPAN	JAPAN	USA	INDIA	INDIA	USA	USA	USA	USA	USA
3	USA	USA	USA	JAPAN	USA	USA	INDIA	VIETNAM	INDIA	INDIA	GERMANY
4	GERMANY	UK	UK	INDIA	JAPAN	JAPAN	JAPAN	JAPAN	JAPAN	BRAZIL	INDIA
5	ITALY	GERMANY	INDIA	UK	TÜRKIYE	AUSTRALIA	VIETNAM	GERMANY	GERMANY	SPAIN	BRAZIL
6	UK	SOUTH AFRICA	GERMANY	GERMANY	GERMANY	TÜRKIYE	AUSTRALIA	AUSTRALIA	BRAZIL	GERMANY	SPAIN
7	ROMANIA	FRANCE	SOUTH KOREA	THAILAND	SOUTH KOREA	GERMANY	SPAIN	SOUTH KOREA	SPAIN	JAPAN	JAPAN
8	INDIA	SOUTH KOREA	AUSTRALIA	SOUTH KOREA	AUSTRALIA	MEXICO	GERMANY	INDIA	AUSTRALIA	POLAND	ITALY
9	GREECE	AUSTRALIA	FRANCE	AUSTRALIA	BRAZIL	SOUTH KOREA	UKRAINE	SPAIN	SOUTH KOREA	AUSTRALIA	POLAND
10	AUSTRALIA	INDIA	CANADA	TÜRKIYE	UK	NETHERLANDS	SOUTH KOREA	NETHERLANDS	POLAND	NETHERLANDS	NETHERLANDS
RANKING EU	2	3	3	4	5	4	2	2	2	2	2
MARKET LEVEL TO ACCESS THE TOP 10											
	792 MW	779 MW	675 MW	818 MW	944 MW	1 621 MW	3 130 MW	3 492 MW	3 710 MW	4 200 MW	4 788 MW

SOURCE IEA PVPS & OTHERS

THE GLOBAL INSTALLED CAPACITY / CONTINUED

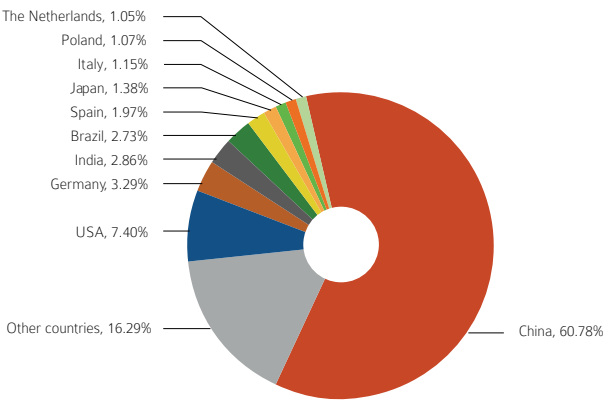
reduces, and therefore the risks. However, the size of the Chinese PV market continues to shape the evolution of the PV market as a whole. As we have seen in 2019, the global growth was limited due to the decline of the first market, which almost wiped out the global growth, while in 2022 and 2023, China’s installations maximized global growth. By mid-2024, approximately 100 GW of new capacity was installed in China – whilst the quarterly volumes are higher than the same period last year, if the 2nd half year is steady this would lead to similar total volumes in 2024 as 2023.

As detailed above, the IEA PVPS choice consists in reporting DC capacities. An estimate of AC capacities would put the new installed capacities number between 305 GWac to 385 GWac in 2023. This number (in the same way as the DC number) is an approximation of the reality and represents an estimated value of the maximum power that all PV systems globally could generate instantaneously, assuming they would all produce at the same time. This number is indicative and should in no case be used for energy production calculation.

Other countries that installed several GW in 2023 and were found in the top 10 countries in the past couldn’t reach the volumes required this year, but remained close: Australia, France and South Korea all installed more than 3 GW in 2023 – all three have had markets fluctuating between 3 GW and 5 GW in the past years, and whilst it appears that France may be accelerating growth, Australia and South Korea have experienced stable markets or slightly decreasing installation rates. Preliminary indications point to a stabilisation in those markets that have reached market competitiveness across most segments as in the USA or Australia, whilst those in protected markets are more closely linked to capacities planned within support mechanisms (France...).

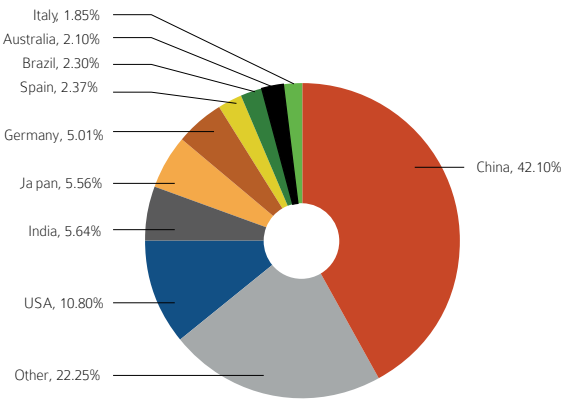
Over 30 countries had more than 1 GW newly installed in 2023, spread across all continents, with the notable arrival of several countries in the Middle East and Africa.

FIGURE 2.5: GLOBAL PV MARKET IN 2023



SOURCE IEA PVPS & OTHERS

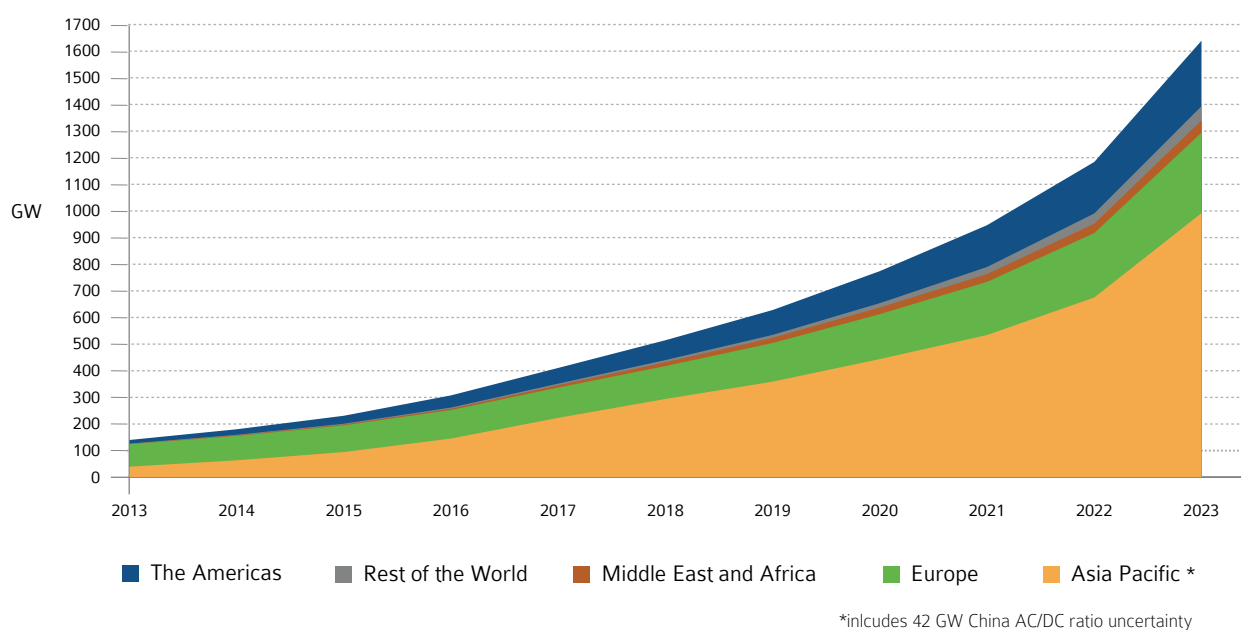
FIGURE 2.6: CUMULATIVE PV CAPACITY END 2023



SOURCE IEA PVPS & OTHERS

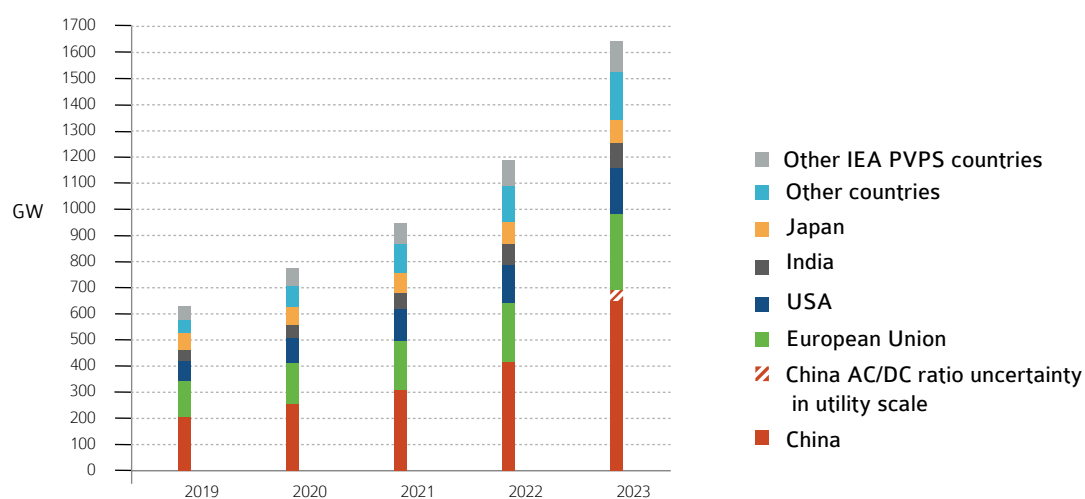


FIGURE 2.7: EVOLUTION OF REGIONAL PV INSTALLATIONS



SOURCE IEA PVPS & OTHERS

FIGURE 2.8: 2018-2023 GROWTH IN MAJOR MARKETS



SOURCE IEA PVPS & OTHERS

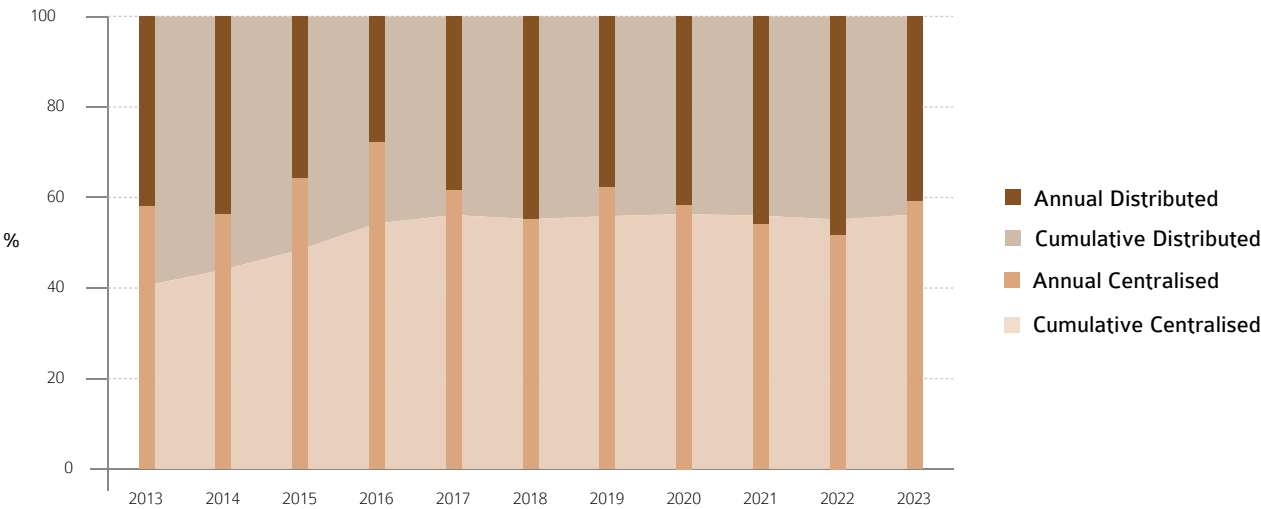
PV MARKET SEGMENTS

We consider three market segments in this report; centralised (large systems above a few MW up to utility-scale and mostly feeding electricity to the grid), distributed (anything below this connected to the grid and connected to a consumption point) and off-grid. Off-grid has been included in distributed volumes in this report’s market statistics With different reporting systems from one country to another, there is no specific system size separating centralised from distributed, however, the segmentation is undertaken in most markets by governments, utilities and industry associations as the impacts are quite different, affecting grid connection, grid capacity, fiscality and taxation and the ability to properly count and evaluate capacities. Additionally, for some markets, the conversion of AC power to DC power is built on different conversion coefficients, in line with industry practices.

Overall, it is estimated that the centralised segment (59% or 267.0 GW) slightly outweighed the distributed segment (41% or 189.0 GW) in new annual capacity in 2023, with a swing towards more centralised systems – although it must be noted that the 42.2 GW of probable extra capacity in China is included in the centralised volumes. This globally balanced result is on the one hand a reflection of the mostly balanced Chinese market but belies significant differences in other major markets.

The share of utility-scale still represented around 56% of cumulative installed capacity, slightly up from last year’s 55%. Off-grid and edge-of-the-grid applications are increasingly integrated into distributed installations, with little ability in most countries to track volumes.

FIGURE 2.9: ANNUAL SHARE OF CENTRALIZED AND DISTRIBUTED GRID-CONNECTED INSTALLATIONS 2013-2023



2

SOURCE IEA PVPS & OTHERS

Except for the European market that incentivized residential segments from the start, most of the major PV developments in emerging PV markets tended to come from utility-scale PV in the past. The drop in module prices and the increasing attractiveness of self-consumption across the world is likely to change this, which has been demonstrated in Brazil and Vietnam, and more recently South Africa in 2023.

UTILITY-SCALE PV

Utility-scale PV plants are in general ground-mounted (or floating) installations. In some cases, they could be used for self-consumption when close to large consumption centres or industries, but generally they feed electricity directly into the grid. Utility-scale applications are thriving in both new and established PV markets. In new or maturing markets, many countries propose tendering processes to select the most competitive projects, or to guide the types of land available.



TABLE 2.2: TOP 10 COUNTRIES FOR CENTRALISED PV INSTALLED IN 2023

COUNTRY	GW
CHINA *	180.89
USA	24.34
INDIA	10.03
SPAIN	6.94
GERMANY	4.65
BRAZIL	4.1
THAILAND	3.48
JAPAN	2.97
NETHERLANDS	2.82
SOUTH KOREA	2.97

*includes 42GW AC/DC conversion uncertain volumes

SOURCE IEA PVPS & OTHERS

TABLE 2.3: TOP 10 COUNTRIES FOR CENTRALISED PV CUMULATIVE CAPACITY IN 2023

COUNTRY	GW
CHINA*	435.26
USA	117.25
INDIA	76.87
JAPAN	36.85
SPAIN	30.64
SOUTH KOREA	23.94
GERMANY	18.39
NETHERLANDS	13.57
BRAZIL	11.49
AUSTRALIA	11.36

*includes 42GW AC/DC conversion uncertain volumes

SOURCE IEA PVPS & OTHERS

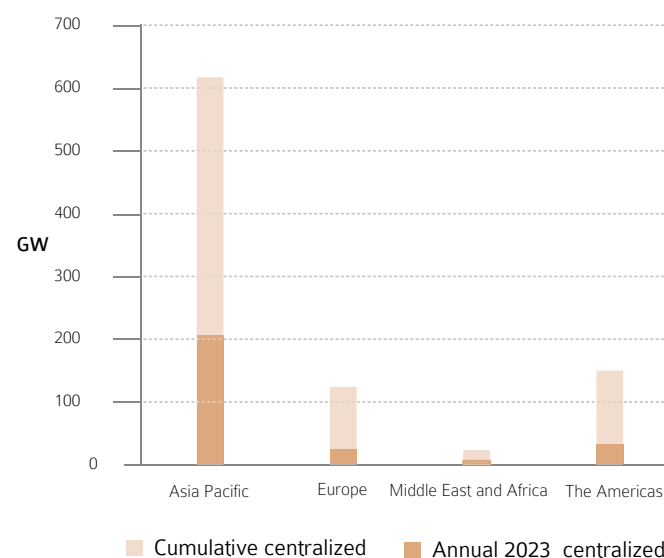
Utility-scale systems are providing a majority of new capacity in some key markets such as India, the USA, Spain and South Korea. The commissioning of very large-scale utility systems in the MENA region – Saudi Arabia and the UAE ahead of a number of other countries in 2024 and 2025 – will add new countries that are driven by centralised systems, however most national markets were not led by this segment in 2023.

Merchant PV, where PV electricity is directly sold to electricity markets and corporate PPAs, where it is directly sold to corporate consumers continues to grow in many countries. Trials conducted

in the past years, especially in 2022 pulled by the high electricity markets costs, gave developers and investors' confidence in the ability to create viable projects. Whilst merchant PV is in its early phases, the impacts of negative prices and price cannibalisation are two subjects that will be closely monitored by investors over the next few years. In parallel, the limitations that are already being seen due to grid congestion, social acceptance or strict environmental impact study requirements remain important barriers. Experience has demonstrated that securing grid connection can lead project developers to tender very low bids just to secure grid connection capacity (Portugal, Spain), whilst some countries have had to specifically invest in and develop grid capacity to ensure the continued development of utility-scale systems (Australia, Austria, Brazil, France).

New utility-scale PV plants are also using trackers to maximise production and the use of bifacial PV modules is increasing relatively fast as well. Floating PV is becoming a significant segment. The addition of storage systems has also become a trend in some countries, either pushed by specific rules in tenders or by attractive conditions for grid services and wholesale markets (Australia, Germany, USA). In 2023, centralised PV amounted to 269.9 GW (59% of new capacity) globally and the total installed capacity for all of these applications amounted to 877.5 GW or 54% of the cumulative installed capacity.

FIGURE 2.10: CENTRALIZED PV INSTALLED CAPACITY PER REGION 2023



SOURCE IEA PVPS & OTHERS

PV MARKET SEGMENTS / CONTINUED

PROSUMERS, POWERING THE DISTRIBUTED PV MARKET

Prosumers are consumers producing part (or all) of their own electricity consumption – and whilst technically any generator in proximity to a consumption point will feed that consumption point, prosumer as a term is reserved for situations where this self-consumption is both based on electron flows and financial flows i.e., the consumer is on the same side of the meter as the generator. The development of prosumer markets is important as it can remove pressure on financial incentives and, in parallel, reduce demand on grid capacity.

Historically driven by simple schemes such as net-metering, prosumer segments are increasingly developed around the concept of self-consumption, where a distinction is made between the electricity consumed on site and the electricity injected into the grid on a close to real time timestep, thereby incentivizing self-consumption.

An important factor in the success of self-consumption schemes is the retail electricity price that is still being maintained artificially low in some countries – even through the high prices of 2022. Subsidies

for fossil fuels are still a reality and reduce the attractiveness of solar PV installations, across all market segments including self-consumption. However, PV markets tend to grow quickly when electricity prices increase, and overall, there is a clear trend toward self-consumption of PV electricity in most of countries, often with regulations offering a value for the excess electricity, either through government mechanisms or utility schemes. This can be done with a FiT, a feed-in-premium added to the spot market price or more complex net-billing including time-of-use rates. Unfortunately, the move towards pure self-consumption schemes can create temporary market slowdowns, especially if the transition is abrupt as consumers and market players adapt their understanding (California, USA in 2023). However, if the market conditions are favourable and the market regains confidence, self-consumption can become a market driver for the distributed segment. Countries where the distributed segment is driving overall market growth include Germany, Brazil, Italy, Poland, Australia, Austria, Sweden and many smaller European countries.

TABLE 2.4: TOP 10 COUNTRIES FOR DISTRIBUTED PV INSTALLED IN 2023

COUNTRY	GW
CHINA	96.29
GERMANY	10.36
USA	9.53
BRAZIL	8.35
ITALY	4.40
POLAND	3.65
JAPAN	3.33
AUSTRALIA	3.15
INDIA	2.99
SOUTH AFRICA	2.60

SOURCE IEA PVPS & OTHERS

TABLE 2.5: TOP 10 COUNTRIES FOR DISTRIBUTED PV TOTAL INSTALLED CAPACITY IN 2023

COUNTRY	GW
CHINA	255.62
GERMANY	63.92
USA	60.10
JAPAN	54.33
BRAZIL	26.33
AUSTRALIA	22.73
ITALY	21.6
INDIA	15.29
POLAND	13.40
FRANCE	12.67

SOURCE IEA PVPS & OTHERS



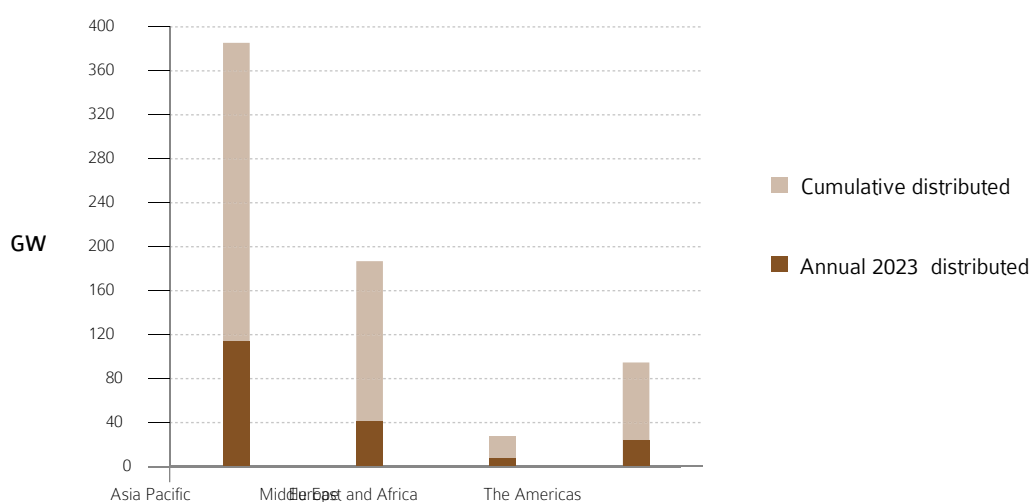
The distributed market has been oscillating around 16–19 GW from 2011 to 2016, until China succeeded in developing its own distributed market: it allowed the distributed PV market to grow significantly by roughly doubling over several years from 2016 to 2018. In 2023, this market grew to 189.0 GW¹, up from 177.7 GW in 2022.

An increasing number of countries, particularly in Europe, promote collective and distributed self-consumption as a new model for residential and commercial electricity customers. This model

allows different consumers located in the same building or private area (collective self-consumption), or in the same geographical area using the public grid (distributed, virtual or delocalized self-consumption), to share the self-generated electricity, unlocking access to self-consumption for a wider range of consumers. If well implemented, this can allow the development of new business models for prosumers, create jobs and local added value while reducing the price of electricity for consumers and energy communities. These models of production could also reduce the impacts on the grid of PV systems by encouraging consumers to adapt their consumption to solar generation hours.

1. Including off-grid systems

FIGURE 2.11: DISTRIBUTED PV INSTALLED CAPACITY PER REGION 2023



SOURCE IEA PVPS & OTHERS

DUAL USAGE AND EMERGING PV MARKET SEGMENTS

The installation of photovoltaic installations on infrastructure or land used for other purpose is a response to the competition for land implied by the massive development of new PV capacities, and the resulting questions around social acceptance. The social acceptance of PV and the availability of land surfaces are key considerations for the massive deployment of new PV capacity in a certain number of countries – typically where there is existing or spreading urbanisation, sustained or regular loss of agricultural land or a strong push to preserve heritage buildings and natural landscapes for aesthetic and cultural reasons.

For ground-mounted installations, competition has arisen between other uses and electricity production, dominated by conflict for agricultural land. By proposing a combined use for both agricultural production and energy production, agrivoltaics offers an acceptable alternative. Floating solar photovoltaics is a particularly dynamic market segment in Asian regions where the tension on land use is strong, using the available surfaces on natural and artificial lakes and reservoirs. These market trends in the centralized segment are strong and underpinned by the same imperative of dual land use. The use of other infrastructure such as canopies on canals or noise barriers along highways is also becoming more common, driven by the same motivating factors of access to land without generating conflict or opposition.

On a smaller individual scale, integrating PV into and onto buildings has been a target for specialised companies for a long time – and has even been the mainstay of public policies for the distributed segment in the past in some countries (most notably France). In particular, the European Union is encouraging member states to develop ambitious targets. It can be attractive because it uses existing buildings but also can lead to a more harmonious integration in the environment, reducing social resistance.

AGRIVOLTAICS

The development of PV on agricultural land has been a reality since the beginning of utility-scale PV deployment. In some cases, crops were replaced by photovoltaics and land use shifted towards electricity production; in other cases, low intensity grazing continued around modules. Agrivoltaics, however, proposes a different approach by using land for both food and energy production at the same time. As PV penetration rates increase in many countries, competition for land can limit PV development – some countries have even regulated access to agricultural land for PV through legislation or conditions in tenders (Italy, France, South Korea).

The potential for PV on agricultural land, and how this segment can

contribute to achieving renewable energy targets has been studied in different regions, and whilst government and developer interest has increased, so has reluctance or opposition from farmers and the general public. To give an example of the potential of this segment, in Japan, a mapping of all agricultural land suitable for PV concluded that just 10% could hold 440 GW of PV – whilst in 2023, Japan's cumulative capacity across all segments is "only" 91.4 GW. South Korea has set a target of 10 GW of agrivoltaics as part of its 46.5 GW 2030 PV target, to be compared with the cumulative capacity in 2023 of 27.6 GW. It has been estimated that covering just 1% of the European Union's agricultural lands could allow the installation of 410 GW – but just 15% of this was installed in 2023 (59.8 GW) whilst the top target of the RePowerEU plan is set at 750 GW in 2030.

PV on agricultural land is implemented with different configurations and designs, and sometimes specific vocabularies:

- PV systems above crops or plants; the PV system allows for growing different kinds of crops but with reduced solar insulation and could provide new services and business models – such as protection against hazards that damage crops (hail, excessive sun), water saving through reduced evaporation, and environments adapted to crops that would not have been possible in the actual or future climate conditions. This dual use requires specific technical solutions such as elevated PV plants, where either the density of the panels is reduced and adapted to the crops needs or the panels are mobile and their position (tilt) is modified to maximize PV production or crop production depending on the weather conditions. The density can be adapted by the design of the PV plant by spreading modules out or by modifying the module itself to be semi-transparent.
- Crops, grassland, pollinator habitat and grazing can be hosted between the rows of PV plants. The systems must enable the land to maintain its agricultural vocation. The shade supplied by the systems can increase grass production and reduce animal (and worker) heat stress – indeed, in some countries increased shading for animals as temperatures increase because of climate change could be an important motivator. The design of the PV plant must be adapted to the activity: adequate space between the rows to allow agricultural machinery, the right height, electrical and dust protection. Ground-mounted PV plants, some with trackers, are implemented at a utility-scale level. Vertical bifacial PV is also being tested in several plants, as the impact of the PV on the land available for the agricultural activity is very low.
- PV systems are also developed and integrated in greenhouses.



Even if the potential for PV on agricultural lands is important, other factors must be taken into consideration. Food production security and agricultural sufficiency are generally the first priority, with the agricultural sector's economic balance, environmental evaluations, social acceptance and water management also important factors. Japan was a pioneer where "solar sharing" was defined in 2003, referring to PV installations where 80% of agricultural yields are maintained - guidelines were published in 2021 and updated in 2023. France, Germany and Italy have published legal frameworks or guidelines in 2022 and 2023. Italy adopted a Decree in December 2023, dedicated to "advanced AgriPV". DIN SPEC 91434, the German Standard for agrivoltaics shapes the German framework and support scheme. Criteria are based on agricultural yield (at least 66% of the reference yield) and the agricultural use of the land must be guaranteed. "Interval" and "overhead systems" are differentiated in the definition, as are the expected outcomes. In France, agrivoltaics is defined by law - an agrivoltaic installation is considered to be an installation which directly contributes to agricultural activities: improving agronomic potential, protection against hazards, improving animal welfare and guaranteeing significant agricultural production.

System costs and profitability vary depending on the importance given to agricultural production compared to energy production. Support mechanisms and financial aid intensity can also vary accordingly. PV systems falling under the most restrictive definition of agrivoltaics typically receive higher incentives.

Generally, in the different frameworks and support mechanisms, two types of PV projects are considered:

- PV plants where some form of agricultural production is maintained. These systems are already economically viable and cost-effective. Energy production dominates but agricultural production is maintained. These projects often participate in the classic competitive tenders or negotiate PPAs.
- PV plants complying with advanced criteria where they enhance agricultural production and farmer revenue. Agricultural production profitability must dominate, and energy production is an added value. This type of plant requires being adapted to the specific crops underneath.

Agrivoltaics is still an emerging market when compared to global PV capacity. Japan has seen more than 1 800 agrivoltaic farms commissioned, but most of them are small systems. Between 2013 and 2021, less than 1 GW was installed. China has also an important installed capacity, but this segment doesn't appear to be monitored

separately. Italy announced support for 1,5 billion EUR including the so-called agrisolar park in rural areas, and by mid-2023 had awarded nearly 0.5 GW of projects. Specific calls for tender have been set for agrivoltaics in numerous countries including Israel with a tender for 100 MW of agrivoltaic systems in 2022. In the USA, there is more than 10 GW of projects that are combined with grazing or pollinator/native habitats, but a much smaller volume (less than 1 GW) protecting crops or on greenhouses.

IIPV: INFRASTRUCTURE INTEGRATED PV

The massive deployment of new PV capacity is presenting new challenges such as social acceptance and the availability of land surfaces for new installations. For ground-mounted installations, competition has intensified between other uses and electricity production; particularly where tensions over land are high: intense urbanisation, need to maintain agricultural production, preservation of buildings and natural landscapes, etc.

Unused land close to transport infrastructures (along roads, highways and railways) and waterways (irrigation canals, banks, dikes, etc.) has significant potential for the installation of PV power

The European Commission's Joint Research Centre (JRC) has assessed the potential for large-scale deployment of vertical solar panels along major roads and railways in Europe. The results reveal a total potential of 403 GWp within the European Union for this type of integration alone, compared to the EU's 2030 target of 750 GWp. The study also establishes that considering only railway lines, the total annual production of PV electricity could potentially reach 250% of the current annual electricity consumption of the EU rail network. This approach offers considerable potential to power the EU's energy needs while contributing to the decarbonisation of the transport sector.

plants and can be an opportunity for new surfaces for PV with better social acceptance, and the rapid decline in PV prices in recent years has allowed PV systems to increase their competitiveness and creates new opportunities for these innovative deployment solutions.

The integration of solar PV energy along transport infrastructure can make a significant contribution to the energy transition. Vertical PV walls along roads, motorways and railways, whether or not integrated with noise barriers, have historically been developed in Europe (Switzerland, Germany, Austria, the Netherlands) and are attracting growing and continuous interest with new projects coming online in the past years around the world (South Korea,

DUAL USAGE AND EMERGING PV MARKET SEGMENTS / CONTINUED

USA, the Netherlands). The electricity production generated by PV installations along roads could contribute to the decarbonisation of this sector as electric vehicles become the norm, by producing energy as close as possible to consumption.

Several pilot projects for elevated PV systems over bicycle paths and roads are announced (Germany, France), as well as experiments on dikes (Netherlands). PV shade canopies on irrigation canals, developed in India since the 2010s, are starting to be announced in regions subject to high evaporation rates (USA, Spain, even France). National railway companies and private developers are also announcing pilot projects along their networks (UK) on previously unexploited linear land.

The types of systems (ground mounted, vertical, integrated into existing infrastructure, in canopies) are numerous and present different levels of technological, industrial and commercial maturity. One of the challenges of these installations lies in the fact that the support's initial use (noise barrier, sunshade) must be preserved while allowing the production of electricity - the proper functioning of structures, systems and equipment, as well as their maintainability, should be ensured. For those installations on linear land (long and narrow) technical and regulatory questions remain to be resolved, in particular with regards to the use of often public lands for private benefits, whilst the electrical architecture and connection possibilities must be studied to achieve both technical and economic viability.

FLOATING PV: CONTINUED GROWTH

In densely populated areas, the proximity of water bodies to load centres is often an advantage. Traditional land-based PV systems face competition for land use with industrial or agricultural activities, or may not be economically viable due to the high cost of land. Japan was one of the early adopters of Floating PV (FPV), with over 200 projects. FPV is even possible in city states such as Singapore, and archipelagos such as Indonesia. The highest installed FPV capacity to date is deployed in China (expected to be over 2.7 GW for about 50 projects by the end of 2023).

By the end of 2023, the global installed capacity of FPV systems is estimated to have reached approximately 7 GW. This represents a significant growth, with the capacity expanding from 5.7 GW in 2022. Most of the growth in 2023 was in Asia-Pacific as in previous years, with large-scale projects in Indonesia (192 MW Cirata Floating Power Plant), Thailand (60 MW), India, Bangladesh (3.2 MW) and China (650 MW). In Europe in 2023, several countries, including Latvia (2.1 MW) and Albania (2 MW), inaugurated their first FPV projects, marking a significant milestone in their renewable energy initiatives. Meanwhile, France (Cintegabelle

8.7 MW) continued to expand its installations. In the rest of the world, significant FPV projects were commissioned in 2023 in Israel (31 MW), Ghana (Bui Hydro-Solar PV Hybrid 5 MW), USA (Canoe Brook Reservoir 8.9 MW) and Colombia (Aguasol 1.5 MW). Project development remained dynamic, with new contracts for increasingly larger projects signed in India after a specific tender round; additionally, aggressive development targets have been set by development companies for the creation of strong localised portfolios of FPV in India (100 MW in Assam), Philippines (100 MW), France (74.3 MW), China (2 GW) or even Zimbabwe (1 GW), whilst local governments and utilities are increasingly incorporating the development of FPV into generation and climate action roadmaps (Germany, Spain, Portugal, India). Specific tenders have been run in some countries (Malaysia, Sri Lanka, Kenya, Portugal).

Floating PV has gained an early foothold on subsidence areas of former (coal) mines and quarries filled with ground water, unsuitable for industrial or agricultural activities and generally with little bioactivities (leading to minimal environmental impacts). Installing FPV on hydropower dams has advantages (for example, streamlining grid connection when conjointly operating the solar and hydro power generation, rather than pure colocation of the FPV plant on the reservoir). Apart from the diurnal cycle (i.e., generating solar power during the day and saving water for hydropower generation at night), there is also a possible seasonal benefit in areas with dry and wet seasons. Depending on the turbines and their reaction times, it is also possible to buffer some of the short-term variability from solar (due to cloud movements) and use the reservoirs as a "giant battery". Planned PV projects on hydropower reservoirs often aim for large capacities such as in Thailand (2.7 GW), Malaysia (2.5 GW) and Zimbabwe (1 GW).

The challenges of near-shore and off-shore marine FPV projects are the more demanding environments, where tidal currents, rich marine life, wind, waves and the presence of salt water all need to be considered. The potential for near-shore areas is significant as unused space can be activated for energy harvesting close to load centres in coastal settlements and harbours. Going further off-shore aggravates the challenges and cost but still has possible applications, especially for powering oil and gas platforms or for using the existing transmission infrastructure and vast ocean spaces between the towers in off-shore wind farms (construction of a 1 GW project started in China in 2023 and pilot projects in the Netherlands and Belgium).

Most of the installations in operation use HDPE plastic floats, (Ciel & Terre, Sungrow, and BayWa r.e. together have a significant market share) but there are an increasing number of different designs, ranging from a combination of floats and metal structures (e.g.



Zimmermann) to membranes that are held in place by large plastic rings (e.g. Ocean Sun). For off-shore applications, more robust designs are being test-bedded, for example by Oceans of Energy or SolarDuck.

Going beyond floating solar, the use of near-shore pile based fixed systems to take advantage of unoccupied space is also being explored, for example with the ambitious goal of 11.25 GW in Shandong Province (China) spread across 10 plants.

The PV market is expected to achieve 20 GW of installed capacity between 2024 and 2030, with India and China contributing to more than 40% of this expansion. Different studies have evaluated regional potentials – a study conducted by researchers in the UK¹ found that covering just 10% of the surface of unprotected lakes and reservoirs within 10 km of population centres worldwide with FPV could generate 1 302 TWh of electricity annually. The study found that on average, countries could cover 16% of their annual electricity consumption with FPV, whilst five countries, including Benin and Rwanda, could cover all their electricity needs this way. Additionally, an Australian study estimates that offshore floating solar panels in favourable conditions could generate 220 000 TWh annually. The fast deployment has not been without problems, as the environmental impacts are still not properly understood, whether it be the biological impacts on water flora and fauna, or hydrological impacts, particularly on flood prone waterways. An increasing understanding may lead to revised development strategies, as seems to be the case in China.

BIPV: A NICHE MARKET

Market perspectives for BIPV are positive, as past obstacles are progressively overcome and the regulatory pressure to decarbonize the energy consumption of buildings increases, particularly in Europe and China. BIPV benefits from the global PV momentum, especially in land-constrained areas where rooftop PV is developing fast. However, the inflationist environment in the USA and Europe heavily hit the construction sector, which hindered the growth of the BIPV market. In China, the real estate sector has known an unprecedented slow-down.

The lack of knowledge amongst professionals of the building construction sector also hinders BIPV growth. Although knowledge is increasing, there is also a real shortage of people combining PV- and building-related skills (across design, performance simulations, knowledge of existing products, technical constraints, ...). Much remains to be done for education and training to generalize BIPV.

Market figures are difficult to estimate. With multiple business

models, different incentives, many kinds of buildings and infrastructure, BIPV covers many applications with a large variety of technical solutions – and what is counted as BIPV also changes from one country to another.

The market is split between industrial products (prefabricated tiles...) and custom-made architectural products, and the differences between custom-made elements and traditional glass-glass modules can be difficult to assess. Simplified BIPV, using conventional PV modules with dedicated mounting structures, is still leading the BIPV market. While supply has been “Western-centered” for a long time, in China many manufacturers are adding BIPV products to their catalogue, including mainstream manufacturers. Research that will benefit BIPV include emerging technologies such as perovskites (for module shape and transparency customization), and cell connections such as matrix shingling for greater freedom customisation.

Depending on the definition considered, the BIPV market ranged from 250 MW to 450 MW per year in Europe last year, indicating a moderate growth or even a stagnation. Globally, the BIPV market probably approached 3 GW, with an increasing number of projects in Asia. It is expected to keep on growing, but at a slower pace than the rest of the PV market.

OFF-GRID MARKET DEVELOPMENT

Numbers for off-grid applications are generally not tracked with the same level of accuracy as grid-connected applications, and volumes are marginal compared to the grid-connected market because the rapid deployment of grid-connected PV and the size of utility-scale -systems. Nevertheless, off-grid applications continue to develop mainly thanks to rural electrification programs essentially in Asia and Africa but also in South America, and mini-grid development in Africa.

In some countries in Asia and in Africa, off-grid systems with back-up represent an alternative to bringing the grid into remote areas or as an anticipation of grid connection. Two types of off-grid systems can be distinguished:

- Mini-grids, also termed as isolated grids, involve small-scale electricity generation with a capacity between 10 kW and 10 MW. This grid uses one or more renewable energy sources (solar, hydro, wind, biomass) to generate electricity and serves a limited number of consumers in isolation from national electricity transmission network. Back-up power can be batteries and/or diesel generators.

1. <https://doi.org/10.1038/s44221-024-00251-4>

DUAL USAGE AND EMERGING PV MARKET SEGMENTS /CONTINUED

- Stand-alone systems, for instance solar home systems (SHS) that are not connected to a central power distribution system and supply power for individual appliances, households or small businesses. Batteries are also used to extend the daily duration of energy use.

PV increasingly represents a competitive alternative to providing electricity in areas where traditional grids have not yet been deployed. In the same way as, mobile phones are connecting people without the traditional lines, PV is expected to leapfrog complex and costly grid infrastructure, especially to reach the “last miles”.

In developing countries (Africa), mini grids are being built to electrify villages and small regions as an alternative to extending costly distribution networks. In some countries, most notably in Australia, as grid infrastructure becomes fragile in the face of extreme climate events (heat waves, fires, floods and storms), micro-grids are being built in edge-of-grid situations to reduce the cost of replacing damaged infrastructure and to provide more resilience for local populations.

Stand-alone systems tend to be specific to countries that have enough solar resources throughout the year to make a PV system viable – regions where there is demand for low energy applications (lighting, smartphone charging). The challenge of providing electricity for lighting and communication, including access to the internet, is being met by PV as one of the most reliable and promising sources of electricity in developing countries. Specific business models are developed (in Africa for instance) and large energy groups are targeting millions of people with such products. After a record 2022 year, it is estimated that 2023 volumes for this type of systems was slightly lower².

PV has also been deployed to power off-grid agricultural purposes such as water pumping installations.

In most developed countries in Europe, Asia or the Americas, there is little need for these systems, however, the future development of off-grid applications for remote islands and rural or underserved areas remains an important potential market.

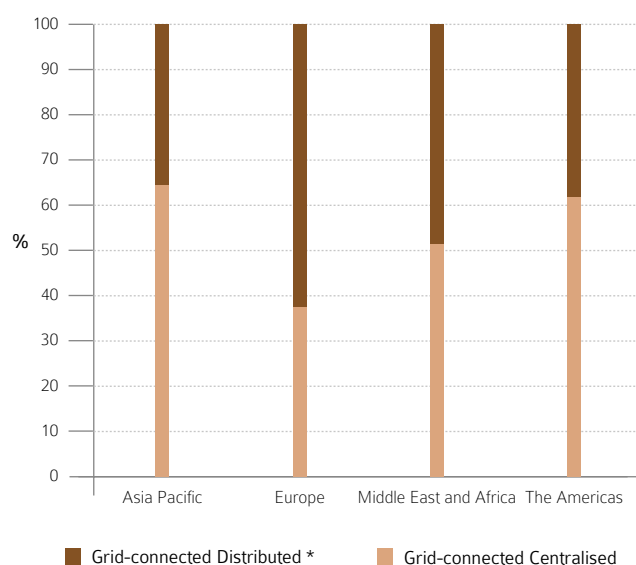
PV DEVELOPMENT PER REGION

The early years of PV development started with the introduction of incentives in Europe, particularly in Germany, and caused a major market uptake in Europe that peaked in 2008. While the global market size grew slowly in the early 2000s, from around 200 MW in 2000 to around 1 GW in 2004, significant investments in Europe pushed the market faster after this. In 2008, Spain fuelled market development while Europe as a whole accounted for more than 80% of the global market until 2010, with 8 GW in 2006, booming to 17 GW in 2010.

From 2011 onward, the share of Asia and the Americas started to grow rapidly as some European markets contracted in a post-boom “bust” phase (Spain, France) with Asia taking the lead. This evolution has had Asia’s share oscillating between 50% and 75%; in 2023, it reached nearly 70%.

Detailed information about most IEA PVPS countries can be found in the yearly National Survey Reports and the Annual Report of the programme. IEA PVPS Task 1 representatives can be contacted for more information about their own individual countries.

FIGURE 2.12: ANNUAL GRID-CONNECTED CENTRALIZED AND DISTRIBUTED PV INSTALLATIONS BY REGION IN 2023



*includes off-grid

SOURCE IEA PVPS & OTHERS

2. <https://www.gogla.org>



THE AMERICAS

The Americas saw 52.6 GW of PV installed in 2023 for a total cumulative capacity of 248.4 GW in 2023. This represented nearly 12% of annual global capacity, down from last year's 16% as the Chinese market leapt beyond other markets capacity to follow. As in 2022, most of these capacities are installed in the USA and Brazil, but several countries have cumulative capacity over the GW level and continue to instal several hundred MW per year (Canada, Chile, Mexico).

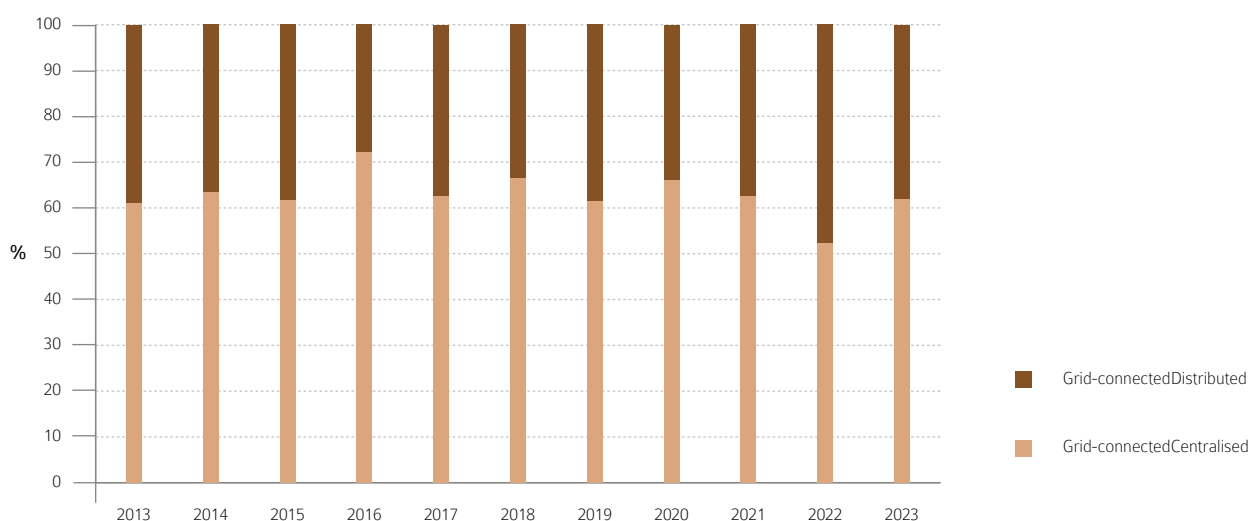
PV is developing in the Americas in both the distributed and centralised segments; the USA is pulled by the utility-scale segment, as is the market in Canada, running both on tenders and PPAs, whilst in Brazil distributed solar is the largest segment. USA has by far the largest installed capacity, adding 33.9 GW to reach a cumulative capacity of 177.3 GW – 2023 was the biggest year for solar in the USA ever, as systems delayed by supply chain and grid queue issues in 2022 were connected. Utility solar jumped from 14.59 GW in 2022 (18.2 GW in 2021) up to 24.3 GW in 2023, whilst annual distributed solar volumes increased as well adding 9.5 GW, up from 8.2 GW in 2022. The Inflation Reduction Act of 2022 (IRA) incentives also played their part in the growth, as did evolving electricity consumption rates and renewable energy standards and pledges as well as support for solar communities. Evolving quite differently, Brazil continued to be the 2nd market in the Americas and the 5th worldwide, adding 12.4 GW of which

67% (8.3 GW) was in the distributed segment – with over 30 GW of distributed projects having requested grid connection, although it remains to be seen how much of this capacity will be effectively commissioned. Its cumulative capacity is now 37.8 GW.

Outside of the IEA PVPS countries, Mexico added 1.6 GW of new capacity in 2023, including the commissioning of the first phase (120 MW) of a 1 GW plant in Sonora. Distributed solar continued strongly with 480 MW of new capacity.

Other South American markets remain low compared to Brazil, but hundreds of MW are still coming online in most countries as smaller centralised system of a few dozen or hundred MW are planned. The PV market picked up in Argentina after a very slow 2022, nearly climbing back to 2021 volume, but was still only a few hundred MW although plans are underway for a multi GW system. In Ecuador, the government selected laureates from its 2021 competitive tenders, but downgraded its election of an initial 345 MW of projects to just 120 MW. Chile was the 2nd largest market in South America, heading towards 10 GW cumulative capacity, with more than 1.3 GW commissioned in 2023, across utility, commercial and distributed solar. In Colombia, approximately 200 MW of utility-scale PV came online in 2023, with more than 1 GW of projects beginning construction, and over 5 GW of capacity was selected in tenders and there is growing interest in distributed solar for commercial projects.

FIGURE 2.13: EVOLUTION OF PV INSTALLATIONS IN THE AMERICAS PER SEGMENT



SOURCE IEA PVPS & OTHERS

PV DEVELOPMENT PER REGION / CONTINUED

ASIA-PACIFIC

The Asia-Pacific region installed 314.4 GW in 2023 and the total installed capacity reached nearly 1 TW, with a minimum of 946.2 GW up to a probable 988.8 GW. The market remained strong across all of Asia Pacific, with a few exceptions where volumes were down compared to 2022 – for example in India, and in the utility-scale segments of Japan and Australia. The region represented 69% of global annual capacity, up from 59% in 2022, and cumulative capacity in the region has now tipped over the 60% mark.

The size of the Chinese PV market makes it a dominant player in the Asian and global PV markets, while all other markets are lagging behind. Asia is home to several IEA-PVPS GW-scale markets in 2023: China, Japan but also Australia, South Korea, with lesser volumes in Malaysia.

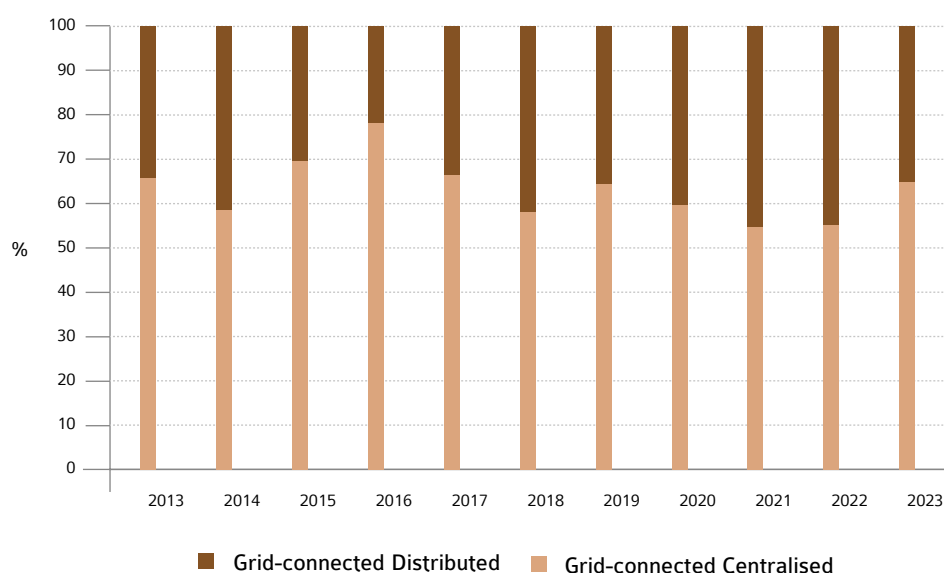
China installed 277.2 GWdc (converted from China's National Energy Administration AC figures using the IEA-PVPS AC/DC conversion ratio as explained in this report), reaching a cumulative capacity of 691.2 GWdc. After two years of balanced growth in the distributed and centralised segments, 2023 saw a real pull from the centralised segment (65% of new capacity). Volumes were spread unevenly across the country, with the highest concentration in the provinces running south down the coast from Beijing, although the capacity in the western province of Xinjiang is catching up to some of the coastal provinces. 2023 was declared a "pivotal year for deployment of renewables" by the National Energy Agency, with the annual target of 160 GW well and truly met.

Outside of the IEA-PVPS network, the largest market in terms of installations and potential is India. Given the population of the country, its potential is on a level with China, (or more, given the need for electrification). After a slow 2020 at just 4.4 GW due to a series of administrative issues and difficulties, the market picked up in 2021 (13 GW) and 2022 (18 GW) and continued in 2023 adding 13.0 GW of grid connected PV (revised from the IEA PVPS Snapshot publication in April 2024). The annual India market share of utility-scale PV was identical in 2023 and 2022 at over 75%. Over 50 GW of tenders were called in 2023, and many of these were for solar+storage, technology neutral or hybrid systems. The distributed market rose slightly but is still far behind the deployment of utility-scale PV; off grid systems represent considerable volumes for a range of applications.

The market in Japan was down to 6.3 GW from 6.6 GW in 2022, reaching a cumulative capacity of 91.4 GW. The Japanese market is slowly contracting year on year, dropping to its lowest annual addition of new capacity since 2012. Tenders through the year were undersubscribed, and the market share of centralised systems dropped to 47% (from 56% in 2022). FiT were increased for some segments to encourage building applied systems, and the distributed market grew for the first time in several years, up 15%.

Both South Korea and Australia had stable markets; South Korea at a steady 3.3 GW (for a cumulative capacity of 27.6 GW) and Australia at 4.2 GW (for a cumulative capacity of 34.5 GW). The two markets are dissimilar though – carried by centralised systems in South Korea and the distributed segment in Australia. Australia is investing in new grid capacity to facilitate further development of utility-scale PV as it heads into what is expected to be regular 100% renewable supply over several hours before the end of 2025.

Chinese Taipei added 2.7 GW or 25% of its cumulative capacity in a single year to reach 10.4 GW, whilst Thailand's market was a standout, adding roughly 4 GW to reach 8.9 GW. Whilst the market in the Philippines remained small in 2023, the liberalised energy market has proved attractive for foreign investment and several GW are in the planning stages, including a 1 GW floating plant whilst construction commenced on a 4 GW system – 2024 should see some significant capacity additions. Malaysia added 0.7 GW for a cumulative total of 3.4 GW. In Pakistan, large volumes of modules (over 7 GW according to some sources) were imported in 2023, with approximately 1.3 GW installed – a series of tenders to equip government buildings were run, as well as a 600 MW competitive tender. Bangladesh added a 275 MW utility-scale system and gave approval for over 1.5 GW as it improved access for foreign ownership. Textile manufacturers have begun equipping premises both in Pakistan and Bangladesh to meet their client's climate targets, in what may be the beginning of a new trend in commercial and industrial systems. In Indonesia, several hundred MW were commissioned, including a 192 MW floating PV system, ahead of a planned expansion to 500 MW.


FIGURE 2.14: EVOLUTION OF PV INSTALLATIONS IN ASIA PACIFIC PER SEGMENT


SOURCE IEA PVPS & OTHERS

EUROPE

Europe led the development of PV for many years, adding a large proportion of the world's capacity through the period 2000 to 2012 to reach 70% of cumulative capacity in 2012. The steep price drops from 2009 to 2012, mostly due to the ramping up of Chinese manufacturing capacity, had significant impacts on European markets. Very fast development of PV over short periods of times ("PV booms") led to a demonstrated opposition from many stakeholders from the traditional energy sector, with different mechanisms resulting in declining markets in several countries. From 2013 to 2017, the growth of European PV installations slowed significantly whilst there was rapid growth in the rest of the world, mainly in Asia and the Americas. In addition, several countries implemented measures to decrease the cost of support mechanisms for PV installations by retroactively changing the remuneration levels or by adding taxes. This phenomenon happened mostly in Europe, where the fast development of PV took place before other regions of the world: Spain, Italy, Czech Republic, Belgium, France and others took some measures with a consequent impact on the confidence of financial backers, developers and prosumers.

The situation improved gradually in most countries and PV installations rose in Europe through the early 2020's. Since then, most European markets have grown each year – especially in the

distributed segment, pulled by growing adoption of residential, commercial and industrial self-consumption sparked by the high electricity prices of 2022. Cumulative capacity across Europe reached 63.3 GW, up in absolute value from 42.4 GW in 2022, but slipping to 14% of the global market, down from 17.5%. The cumulative capacity in Europe is now 305.2 GW.

It is important to distinguish the European Union and its countries, which benefit from a common regulatory framework for part of the energy market, and other European countries which have their own energy regulations and are not part of the European Union.

Whilst most European countries used Feed-in Tariff schemes to start developing PV, the movement to self-consumption (or variants) for distributed PV is accelerating while tenders and PPA's become the standard for utility-scale PV. These trends are not unique to Europe, but self-consumption developed faster here than in other locations, no doubt due to the high electricity consumption prices. The development of collective and delocalized self-consumption is also accelerating in EU countries where regulatory frameworks are catching up to market demands¹.

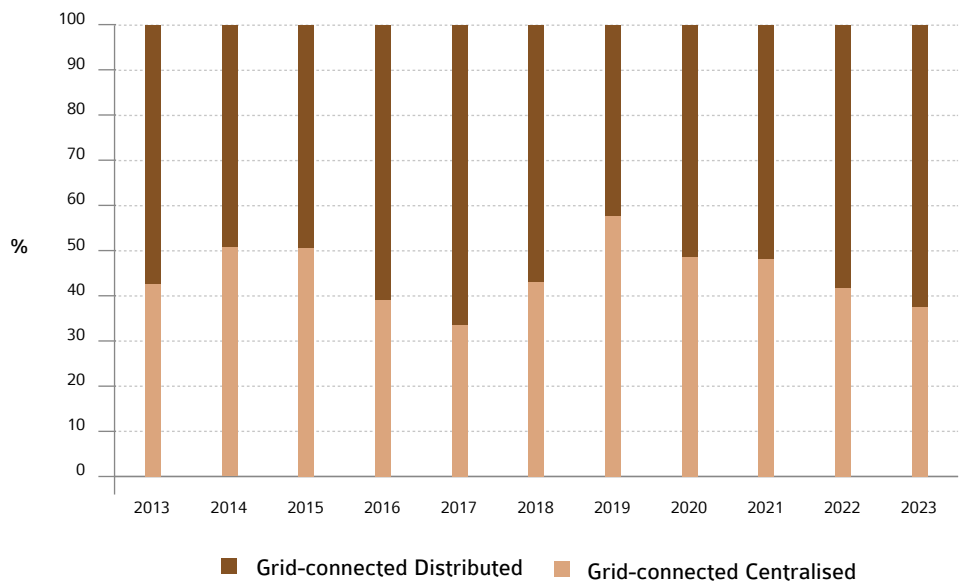
1. See the IEA PVPS Task 14 paper "Self-consumption of electricity produced with photovoltaic systems in apartment buildings - Update of the situation in various IEA PVPS countries"

PV DEVELOPMENT PER REGION / CONTINUED

BIPV has been incentivized in Europe more than in any other part of the world in the past but remains a niche market after several GW of installations. Simplified BIPV seems to develop well in some countries and is likely to increase, always as a niche market, with a slow deployment by different countries of mandatory solar of some sort in building regulations as a response to ambitious climate

targets. Merchant utility-scale PV developed in Spain, Germany and France initially with more recent projects in Romania, - and could take a significant market share in a near future, along with PPA and corporate PPA's whose market shares continue to grow. In general, PV development in Europe continued to be dynamic over 2023 as China's manufacturing overcapacity is targeting the more open European market, rather than the USA and India.

FIGURE 2.15: EVOLUTION OF PV INSTALLATIONS IN EUROPE PER SEGMENT



SOURCE IEA PVPS & OTHERS

EUROPEAN UNION

At the end of 2023, the total installed PV power capacity in the European Union had reached 273.7 GW, or 90% of the capacity in Europe. As in the wider European market, residential, commercial and industrial rooftops are the most important segment, with 61% of new capacity additions (up from 57% in 2022). The market in 2023 was largely influenced by the remnants of the electricity market price spikes of late 2022 and the aggressive pricing of modules as manufacturing capacity in China increased significantly. Fourteen countries installed more than 1 GW, with Germany heading the list (15.0 GW), followed by Spain (8.9GW), Italy (5.3 GW), Poland (4.9 GW), the Netherlands (4.8 GW), France (4.0 GW) and Austria (2.6 GW). Belgium, Sweden, Hungary, Greece, Portugal, Bulgaria and Romania all installed between 1 GW and 2 GW of new capacity.

Market split between centralised and distributed generation varies from country to country; the European markets with more than 0.5 GW centralised PV where utility-scale PV is a market driver are Spain, the Netherlands, Hungary and Bulgaria; on the opposite end, more than a dozen countries with over 0.5 GW of new distributed capacity were driven by the distributed market – most notably, Germany, Italy, Poland, France and Austria.

The Netherlands continued to lead in terms of installed capacity per capita in the EU with 1288 Wp. The number of EU countries that had a penetration rate above the European Union average, now 488 Wp/capita, doubled from six to twelve – with five above 700 Wp per capita: Germany (974 Wp), Belgium (842 Wp), Spain (806 Wp) and Austria (700 Wp) and an additional five above the EU average.



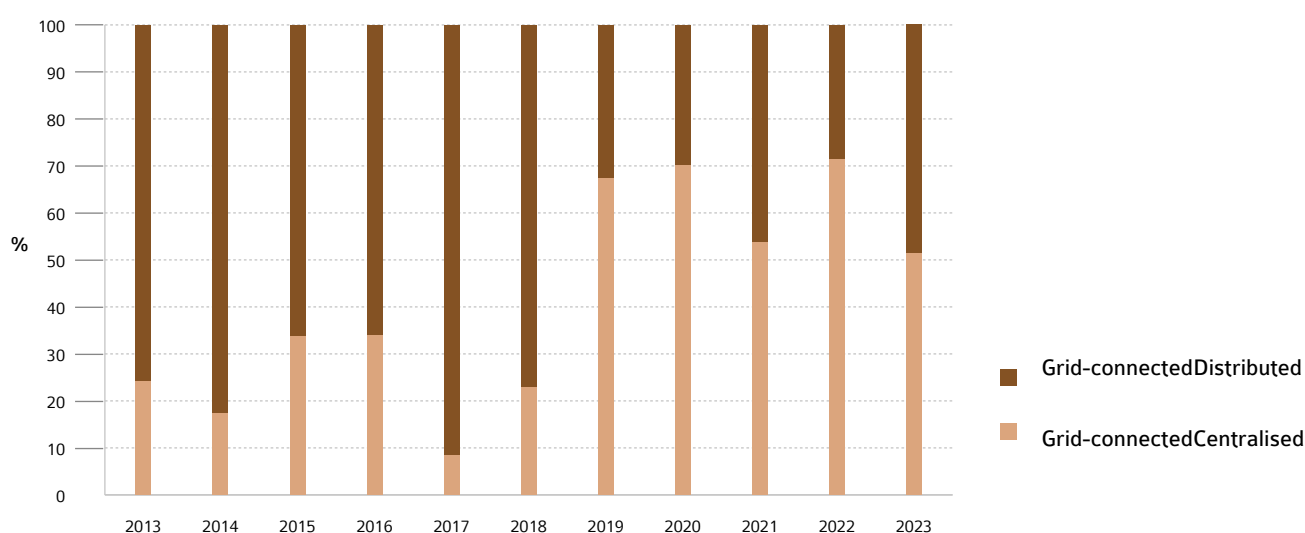
OTHER EUROPEAN COUNTRIES

Outside of the IEA-PVPS network, UK was a growth market with 1.3 GW in 2023, nearly double the 2022 value. The country had 16.0 GW of PV at the end of the year 2023 and awarded 1.9 GW of capacity in tenders last year, expected to come online through 2025 to 2028, pushing the utility segment to the front. Switzerland continued to expand with 1.6 GW installed, once again, almost all rooftop systems.

MIDDLE EAST AND AFRICA

In Middle East and Africa countries, the development of PV remains modest compared to the larger markets, especially in the African countries. However, almost all countries saw a small development of PV in the last years and some of them a significant increase. There is a clear trend in most countries to include PV in energy planning, to set national targets and to prepare the regulatory framework to accommodate PV.

FIGURE 2.16: EVOLUTION OF PV INSTALLATIONS IN AFRICA AND THE MIDDLE EAST PER SEGMENT



SOURCE IEA PVPS & OTHERS

MIDDLE EAST

With high irradiation, the Middle East is amongst one of the most competitive places for PV installations, with PPAs granted through tendering processes among the lowest in the world, building on excellent irradiation levels. In 2023, several large-scale systems came online, bringing the annual capacity up to 7.5 GW, well above last year's 5.5 GW. Three countries in the region added 1.9 GW in 2023: Saudi Arabia (for a cumulative total of just 2.9 GW), Türkiye, who reached the region's highest cumulative capacity of 14.4 GW, and the UAE (cumulative capacity of 7.1 GW). Israel maintained its market at 1.1 GW for a total of 5.7 GW, with the other regional markets trailing.

In the region, energy prices are often supported by government spending, which limited the ability of PV to compete for years, however, countries such as Iran, Qatar, Kuwait, Saudi Arabia,

Bahrain, Jordan, Oman and the United Arab Emirates have or are defining targets for renewable and solar energy for the coming years. Conditions are slowly changing for distributed PV, with net-metering being proposed in some countries such as in Türkiye, Egypt, Dubai where the upper limit was lowered in 2023, Bahrain, Jordan (transitioning to net billing in 2024), and FiT or other mechanisms in Iran, Israel, Saudi Arabia, Tunisia, with plans to introduce similar schemes in Qatar, and Morocco. The UAE is facilitating the connection of distributed generation to the network to help relieve peak loads. Türkiye is looking to sustainable building mandates to promote distributed solar. Another trend in the fast-developing region is the willingness for governments to develop brand new cities or neighbourhoods, which aim at becoming showcases of renewable energies. This was the case for Masdar City (UAE) or Spark and Neom (Saudi Arabia).

PV DEVELOPMENT PER REGION / CONTINUED

For centralised PV, tenders are an integral part of the plans for PV development in the short and long term in the region, with government and state-owned organisations tendering for single site or multi-site projects as procurement exercises. Examples in 2023 include Saudi Arabia (3.7 GW in the 5th round of its NREP). Morocco called for 400 MW of solar+storage for Noor III and Tunisia called tenders for 1 GW. Projects for solar powered green ammonia or hydrogen are underway in Jordan, Egypt and the UAE (the UAE is looking to green the steel industry). In parallel, plans for a 10 GW metallurgical silicon, polysilicon, cell and module manufacturing plant in Saudi Arabia have been initiated.

AFRICA

The African market is dynamic but difficult to follow, with market reports diverging in terms of capacity, depending on their sources. On top of weak reporting standards and capabilities in many African countries, probably significant volumes of off grid micro and small-scale PV seem to be unreported to authorities. COVID impacted growth in Africa, and the shrunken market from 2019 to 2021 resulted in a break in the previous positive trend of PV development and rural electrification, as fiscal resources were diverted to food and primary necessities.

Africa is by far the smallest regional market, with just an estimated 20 GW of cumulative capacity installed, of which over 3.7 GW was installed in 2023¹. South Africa is the largest market, with 3.0 GW of new capacity in 2023 – of which 2.6 GW was distributed – to reach a cumulative capacity of 7.5 GW. Whilst other countries have nascent commercial and industrial segments, volumes remain low – the 30 MW Gorou Banda Solar Power plant was inaugurated in Niger, Zambia saw a 33 MW plant come online with plans for over 2 GW more, and tenders for a 30 MW solar+storage plant in Eritrea and 30 MW floating PV in Kenya were launched. Financing for a 50 MW system in Sierra Leone was confirmed.

Large-scale systems remain a main contributor, but the commercial and industrial segment is slowly gaining momentum and maturity as companies seek to hedge against future rises in electricity prices by generating their own power.

Large scale systems for low-carbon hydrogen projects are underway or under discussion, primarily looking to produce ammonia for fertiliser, both strengthening Africa's food security and taking advantage of the excellent irradiation available in many African countries. Experts are confident that the production of highly competitive H2 can be reached.

1. According to the African Solar Industry Association.



The question of African power infrastructure and markets is essential since many countries have a small, centralized power demand, sometimes below 500 MW. In this respect, the question is not only to connect PV to the grid but also to reinforce the electricity grid infrastructure and interconnection with neighbouring countries.

Mini-grids are an important tool for rural electrification, and a series of projects were announced or developed in 2023 (Angola, Botswana, Ghana, Nigeria, Senegal, Somalia, for example). This segment is still very dependent on grants and subsidies while it tries to identify solutions that will lead to pure commercial bankability.

One of the main difficulties for these segments is attracting private investment (with investors wary of the risk levels and upfront cost). However, new renewable energy markets are showing greater appeal to international and local investors - and the most competitive segment for the development of solar in Africa, especially in remote areas i.e. PV plants to replace or complement existing diesel generators, is of interest to investors.

Early solar PV development was supported by government and donor-supported rural electrification with micro and small systems, but the transition to a more market-based development can now be observed. Pay-as-you-go models are used to bypass financing difficulties for residential consumers whilst different pricing formats exist to foster access to clean and reliable electricity. Off-grid PV applications, such as water pumping, are expected to play a growing role in bringing affordable power to consumers.

Support policies and facilitating regulation are more than ever necessary to accelerate uptake ahead of the expected increase in energy consumption and electricity access due to a multitude of factors including population growth and socioeconomic dynamics (increases in the energy consumption per capita for those countries with high electrification rates), decarbonisation of electricity production, electrification of industry and transport, production of green hydrogen...

Africa is already facing severe climate change consequences including water stress and reduced food production, so emerging dual usages such as agrivoltaics and floating PV are potential tools to increase food resources and protect water resources whilst producing electricity.

PV DEVELOPMENT PER REGION / CONTINUED

TABLE 2.6: 2023 PV MARKETS STATISTICS IN DETAIL

COUNTRY	2023 ANNUAL CAPACITY (MW)			2023 CUMULATIVE CAPACITY (MW)		
	DECENTRALIZED	CENTRALIZED	TOTAL	DECENTRALIZED	CENTRALIZED	TOTAL
AUSTRALIA	3 182	972	4 153	23 165	11 356	34 522
AUSTRIA	2 295	308	2 603	5 832	563	6 395
BELGIUM	1 621	185	1 806	9 533	422	9 955
CANADA	99	724	823	2 337	5 003	7 340
CHINA	96 290	180 886	277 176	255 980	435 261	691 241
DENMARK	321	167	487	2 031	1 879	3 910
FINLAND	302	16	318	959	49	1 008
FRANCE	2 587	1 374	3 961	12 701	10 963	23 664
GERMANY	10 357	4 648	15 005	63 916	18 389	82 305
ISRAEL	830	320	1 150	3 472	2 185	5 657
ITALY	4 399	856	5 255	21 060	9 259	30 319
JAPAN	3 328	2 972	6 300	54 517	36 849	91 366
SOUTH KOREA	514	2 792	3 306	3 675	23 944	27 619
MALAYSIA	266	444	710	1 289	2 100	3 390
MOROCCO	693	35	728	1 679	35	1 714
NETHERLANDS	1 964	2 824	4 788	9 463	13 574	23 037
NORWAY	299	4	303	653	4	657
PORTUGAL	775	509	1 285	1 522	2 300	3 822
SOUTH AFRICA	2 600	365	2 965	5 205	2 299	7 504
SPAIN	2 047	6 940	8 987	8 346	30 640	38 986
SWEDEN	1 611	77	1 688	3 945	270	4 215
SWITZERLAND	1 641	0	1 641	6 306	68	6 375
THAILAND	1 103	3 484	4 587	2 380	6 620	9 002
TÜRKIYE	1 027	840	1 867	11 309	3 084	14 393
USA	9 530	24 345	33 875	60 096	117 248	177 344
IEA PVPS	149 681	236 086	385 766	571 372	734 366	1 305 739
BRAZIL	8 346	4 100	12 446	26 330	11 487	37 817
INDIA	2 995	10 025	13 020	15 771	106	92 645
NON-IEA PVPS	39 346	30 890	70 244	156 529	179 941	336 218
TOTAL	189 027	266 975	456 010	727 900	914 307	1 641 956

SOURCE IEA PVPS & OTHERS



three

POLICY FRAMEWORK

As the cost environment of PV changed over the past two years, from the high costs yet high competitiveness of mid-2022 to the plummeting module costs but lesser competitiveness of late 2023, the importance of policy support even in conditions of market competitiveness has become increasingly apparent.

From the late 90's onwards, the development of PV was pushed by various support mechanisms – from feed-in tariffs, direct subsidies, and tax credits to competitive calls for tender and feed-in premiums. The initial goal was generally to compensate for the lack of competitiveness by reducing the gap between the cost of electricity from PV and the cost of electricity from conventional sources. More recently, the rapid reduction in PV costs has meant that competitiveness is no longer a problem in several countries and across different segments (for more detail, see Chapter 6, competitiveness of PV electricity).

Support mechanisms for centralized PV are increasingly becoming additional to market remuneration with competitive tenders operating as guarantees for a minimal remuneration after taking into account market sales, pushing the development of power purchase agreements (PPA) and market sales (merchant PV). These Contract for Difference schemes demonstrated their ability to funnel profits from high electricity market costs back to the government in 2022. When PV is so competitive that no financial incentives are required, large-scale systems are moving directly to PPAs, with a trend to diminish contract length – inevitably, many PV systems operating initially in these private contracts will end up with electricity sales on wholesale markets (merchant PV). After a series of undersubscribed tenders in 2022 due to the attractively high market electricity prices of 2022, tenders in some European countries were largely oversubscribed in 2023 as the stability of guaranteed remuneration and (easier) financing was balanced with

volatile electricity market costs and higher financing costs.

In new markets, support for distributed PV often begins with net metering, before setting feed-in tariffs, and finally ends in some form of net billing where excess electricity after self-consumption is bought at a utility or government-set rate; this rate can be lower than wholesale electricity costs (to discourage excess) or benefit from feed-in tariffs at premium rates. Direct subsidies and tax credits remain present across the world, although direct subsidy policies tend to be fragile due to their high upfront costs for governments – and are increasingly reserved for stimulating specific market segments.

For residential and commercial markets, in many countries self-consumption has become the model of choice with regulations and support measures being adapted to accommodate this. Whilst in some competitive markets support mechanisms have been stopped or adapted to encourage emerging segments such as building Integrated PV or AgriPV, climate imperatives, the search for energy sovereignty, and ambitious PV development goals mean that other countries are stepping up to accelerate deployment by increasing support budgets or support levels. Many different indirect policies to encourage or facilitate PV such as mandatory solar on buildings and car parks and support mechanisms addressing permitting complexity and costs, facilitated access to electricity markets, or grid access policies for prosumers are being used across the world to accelerate PV deployment.

Where complete competitiveness is not yet present, support schemes are evolving according to market maturity, and investor confidence. Therefore, targeted financial incentives might still be needed for some years to overcome costs or investment barriers in specific countries. The electricity market volatility of the past few years

has demonstrated that support mechanisms need a certain level of agility, and that competitiveness can be insufficient for financing long-term project viability.

Where market volumes are strong, a large part of new policies are focused on self-consumption schemes, citizen communities, and innovative forms of collective and delocalized self-consumption. Policies supporting self-consumption might be considered as non-financial incentives since they set up the regulatory environment to allow consumers to become prosumers or an energy community.

The electrification of usages is an important factor in the energy transition, and the sale of battery storage, heat pumps and EV are increasing rapidly for residential and commercial users, whilst big batteries are enabling fast transitions in areas with high penetration rates of renewable energies and increasingly saturated grids. In parallel, the development of “green” hydrogen projects powered by solar and other renewable electricity continues stronger than before as the fragility of gas supply pushes governments to accelerate support for alternatives.

Taxes and the financing of distribution and transmission grids are still animating the debate, shaping the regulatory framework and impacting the business models and the price for PV electricity.

PV MARKET DRIVERS AND SUPPORT SCHEMES

The question of market drivers is a complex one since the market is always driven by a combination of several regulations and incentives.

ACCESS TO SUPPORT MECHANISMS AND REMUNERATION MODELS

Access to support mechanisms can be through open-access schemes (either unlimited or capped in volume or time, generally with a series of mandatory requirements relating to system size or installation type, sometimes associated with minimal installer qualifications or product certification) or through competitive tenders. Remuneration models tend to include feed-in tariffs (FiT) or premiums (FiP)—sometimes for all the electricity generated from a system, sometimes for excess generation after self-consumption (net billing), green certificates or direct subsidies in open access schemes, whilst tenders are seeing a shift away from feed-in tariffs to feed-in premiums and contracts for difference (CfD) – but also to tenders for electricity or generation capacity procurement. Power purchase agreements between generators and consumers can be modelled on any of these mechanisms.

FEED-IN TARIFFS AND PREMIUMS IN OPEN ACCESS SCHEMES

Predefined feed-in tariffs remain an important tool for ensuring roll-out of PV in the distributed segment, and whilst many existing schemes continued through 2023, others were discontinued. Feed-in tariffs work on a simple principle – electricity produced by the PV system and injected into the grid is paid at a predefined price and guaranteed during a fixed period (often 10 or 20 years). FiT are paid in general by official bodies or utilities and were set-up to stimulate local PV market segments. Generally, only available for small or small to medium-sized systems, the FiT can be fixed over the contract period, or be indexed to inflation or some other indicator, and it can be made available for systems that inject the entirety of their electricity into the grid or, increasingly, only the excess after self-consumption (also called net billing). In some countries, system owners may be able to choose between total injection or the injection of excess after self-consumption and may even be allowed to migrate from one model to another. FiT have demonstrated their efficiency as drivers for the development of residential markets and remain a tool for incentivising self-consumption by managing excess generation without batteries.

Amongst the IEA PVPS member countries that had a government (national, federal or state) mandated open access FiT scheme in 2023, most cover the residential sector, (Austria, Canada, China (ended), France, Germany, Greece (new), Japan, Portugal, Switzerland, Thailand, USA), sometimes extending to the commercial and industrial segments (Austria, France, Germany, Japan). With increased competitiveness, a few countries have or are phasing out government-based FiT schemes (Australia, where utilities set net billing feed-in tariffs as part of their customer acquisition and loyalty strategies, some states have set minimum rates). With the large drop in price experienced over 2023 for modules, some countries (France, California, New York and Indiana in the USA), once again adjusted feed-in tariff levels to take into account the changing profitability of PV.

Increasingly countries are either dropping FiT (in 2023, South Korea) or reserving them for excess injections after self-consumption (net billing FiT) as grid parity is reached. Depending on the country specifics, FiT can be defined at the national level and at the regional, county or city level (Australia, Canada, China, USA etc.) with some regions opting for it and others not, or with different characteristics. FiT can also be granted by utilities themselves (Australia, USA, China (Hong Kong)), outside of the policy framework to increase customer fidelity.

Feed-in premiums (FiP) are premiums paid on top of the wholesale electricity market price. Fixed and variable premiums can be considered. Sweden and Austria are using a fixed FiP for small decentralized systems – in the case of Sweden, it is managed through tax credits per kWh generated.

Defining FiT or FiP levels that adequately incentivise PV without



overcompensating can be a delicate task, particularly when costs are volatile or subject to steep declines. Entities bearing the cost of FiTs (governments or utilities) will generally seek adjustment mechanisms to ensure that market booms do not lead to cost blowouts and/or over-compensation, a lesson learned from past market booms that occurred in countries such as Spain in 2008, France in 2009, Czech Republic in 2010, Italy in 2011, Belgium in 2012, to a certain extent in China in 2015, 2016 and 2017, and to a lesser extent to other countries. These booms strained budgets and negatively affected the public perception of PV, and most of these markets took years to recover and reexperience growth.

To keep market development stable or financially viable for cost bearers, adjustment mechanisms can include periodic industry negotiations, inflation, and market growth indexation.

Many countries adopted the principle of decreasing FiT levels over time or introduced limited budgets, building on the experience of a consistently reducing price curve for PV systems. In Germany, the level of the FiT can be adapted monthly to reduce the profitability of PV investments if the market is growing faster than the target decided by the government. In Japan FiT are adjusted regularly – and in 2023 were increased for commercial-sized rooftop systems to encourage this specific segment. In France, the FiT decrease is dependent on both installation rates and economic indicators. The economic indicators and government intervention also allow for increased or decreased FiT if economic conditions (such as cost increases or booming investment) require it.

INCENTIVIZED SELF-CONSUMPTION

Self-consumption, supported by different mechanisms such as net-metering, net-billing, premiums on self-consumed electricity or investment subsidies is becoming the norm in an increasing number of countries. Various forms of support to self-consumption schemes exist, from the historically deployed “net metering” used to develop the market of small-scale PV installations on buildings that were adopted in a large number of countries, although with different definitions of what, precisely, net metering meant (the same vocabulary can imply different regulations and different remuneration models. The best example is in the USA, with the wording “net-metering” being used for different self-consumption schemes in different states).

Genuine “net-metering” which offers credits at the same rate as consumption for PV electricity injected into the grid, has previously supported market development in Belgium, Canada, Denmark, Hungary, the Netherlands, Portugal, South Korea, the USA, and even Pakistan but such policies continue being replaced by net billing, self-consumption policies encouraging real-time consumption of PV electricity (Poland, Arkansas, North Carolina, Idaho, and Hawaii in the USA in 2023 alone).

Net billing can be incentivized with a feed-in tariff (or feed-in premium added on top of the spot price) for the excess PV electricity fed into the grid. This is for example the case in France,

Italy, Japan. It can also be associated with time of use rates or low rates to discourage sending excess electricity to the grid, for example encouraging investment in battery storage (Australia and California (USA)). As grids in some parts of the world become more congested, or as distributed solar outpaces consumption, curtailment policies are put in place that no longer guarantee access to the network at all times (Australia, South Korea).

Although net metering is being replaced in historical markets, countries such as Malaysia or Ecuador introduced net metering for residential PV owners recently. Several emerging PV countries have implemented net metering schemes in recent years (Chile, Jordan, UAE (Dubai), Morocco and Tunisia). While the self-consumption and net metering schemes are based on an energy compensation of electricity flows, other systems exist. Italy attributes different prices to consumed electricity and the electricity fed into the grid.

DIRECT SUBSIDIES, REBATES AND TAX BREAKS

Direct subsidies remain in place in many countries. Most of the time they cover only a part of the total installation cost. PV is characterized by limited maintenance costs, no fuel costs but high upfront investment and this has led some countries to put policies in place that reduce the upfront investment to incentivize PV. Direct subsidies were implemented in the early phase of PV development in countries such as Austria, Australia, Canada, Finland, Italy, Japan, South Korea, Lithuania, Norway, and Sweden just to mention a few.

Incentives can be granted by a wide variety of authorities or sometimes by utilities themselves. They can be unique or add up to each other. Their lifetime is generally quite short, with frequent policy changes, at least to adapt the financial parameters to current project economics and political priorities. Next to central governments, regional states or provinces can propose either the main incentive or some additional ones. Municipalities are more and more involved in renewable energy development and can offer additional advantages. In some cases, utilities are proposing specific deployment schemes to their own customers, generally in the absence of national or local incentives, but sometimes to complement them.

In most countries, this support mechanism has not demonstrated its ability to support and accelerate PV development over the long term and was progressively replaced by FiTs, although not completely phased out everywhere. However, because of the psychological attractiveness of direct upfront subsidies for the residential sector (depending on local culture they can be more attractive to end users than deferred payments through FiT), some countries have continued to use direct subsidies alone (Flanders in Belgium, South Korea, Norway, Poland, Italy, Switzerland) or in conjunction with other support mechanisms (France, for self-consumption with net billing), Australia, Austria, Germany, Greece for PV with storage. Switzerland gives bonuses on top of its direct subsidies depending on the size, inclination, and altitude of systems. South Korea continues to provide direct subsidies in different programs for

PV MARKET DRIVERS AND SUPPORT SCHEMES / CONTINUED

residential systems (since 2004) and building applied systems on commercial buildings (with priority given to BIPV systems).

Governments have also used direct subsidies to encourage very specific segments on an experimental basis – Japan joined Austria, Finland with subsidies for agricultural systems and sometimes systems on other specific land usages. The USA has a diversity of state and local subsidies for solar installation, often targeted at increasing deployment in low-income and/or historically marginalized communities (South Korea also). The landmark Inflation Reduction Act also included increases to grant programs targeting both rural deployment and deployment in indigenous communities, as well as creating several new grant programs incentivizing deployment in historically marginalized communities. Different regions in Italy have set up capital investment subsidies in 2023; one region has launched a scheme that should be self-sustaining for families in energy poverty situations that cover installation and maintenance costs, with self-consumed electricity free for the residents and excess electricity sales feeding the financing fund. Many countries have capital subsidies for local government authorities installing PV (Italy, Japan South Korea, some states in the USA, California, New York, Nevada, Illinois, etc) – often with conditions mandating the use of building integrated PV.

Tax credits have been used for a long time around the world, either on their own or associated with FiT, direct subsidies or rebates. (as early as 2005 in France, USA), and spread to a large variety of countries, ranging from Belgium and Canada to Japan and others. Tax credits can be applied to equipment costs, labour costs or, more rarely, even electricity (Sweden).

Tax credits have been introduced (or re-introduced) in the past few years including in Germany (2022), Italy (2023), Sweden (2015, 2021). Tax credits can be open to individuals (USA, Italy, Switzerland, Germany, Finland, Sweden) or commercial entities (USA, Switzerland, Germany, Sweden (limited to small systems)), and operate on a yearly or multi-year basis. Tax credits remain a popular tool that is relatively easy to adjust (for example in the USA in 2022 as part of the Inflation Reduction Act). Storage installed with, or added to existing PV can also be eligible for tax credits (USA, Austria and Sweden). Occasionally, tax incentives can be increased for low-income households (USA, UK) or limited to self-consumption systems (Spain), and for specific requirements such as local module production (Italy, USA).

Elsewhere, exemption from import duties for PV modules can also stimulate markets with little or no local manufacturing (Pakistan, Benin, Uganda, Mali, amongst others).

COMPETITIVE TENDERS

Competitive tenders are used to control the volume, budget, location and type of photovoltaic systems that could benefit from support mechanisms. Competitive tenders have been adopted in many countries around the world, with the clear aim of increasing the competitiveness of PV electricity. They can be run by state organizations or utilities looking to secure increased supply and security of supply on one hand and reduce supply costs on the other.

In Europe the Netherlands and France were early adopters of competitive tenders, running them as early as 2011/2012. By 2018, uptake was more widespread with roughly 10 countries running tenders (including Germany, Poland), doubling up to at least 20 European countries having trialed or validated tender schemes since then, from Scandinavian countries to the Baltic states down to Mediterranean rim countries. In the past 5 years, large volumes were awarded in tenders in Germany, Italy, Poland, the Netherlands, the UK, Spain and France

Their use worldwide is increasingly used to stimulate and regulate medium to large-scale systems across the world, either tendering access to support mechanisms, and land or electricity procurement contracts. They are also used to stimulate investment in specific types of systems, for example floating PV or, more commonly in the past few years, agrivoltaics. A European Commission report “on the performance of support for electricity from renewable sources granted by means of tendering procedures in the Union” published in November 2022 concludes “that introduction of tenders for renewables was a clear success for the European Union” and that they “reduced the support cost significantly compared to administrative schemes, enhanced the deployment of renewable capacities and provided a solid framework for technological improvement”.

Initially, the first remuneration models in tenders were feed-in tariffs (FiT) where remuneration is based on a fixed price per kWh, and whilst they are still used in some countries for some segments, particularly commercial and industrial systems on buildings (Japan, Italy, Egypt), governments are steadily replacing them with feed in premiums or with Contract for Difference (CfD). A Feed-in Premium (FiP) is a fixed premium that is paid on top of the variable electricity wholesale market price, leading to variable remuneration. In 2023, Austria, transitioned their tendered feed-in tariffs to feed-in premiums, joining Germany, the Netherlands who used this mechanism already. Japan uses both FiT and FiP in its tenders, depending on system size. A Contract for Difference scheme (UK, Greece, France, Italy, Hungary, Australia and Spain) is a FiP that ensures constant remuneration by covering the difference between the expected remuneration and the electricity



market price. Tenders have also been run with some or all of the remuneration based on green certificates (Australia) or some other form of “green” bonus on top of market remuneration (Flanders, Belgium).

In March 2023, the European Commission’s proposed revision of the EU’s internal electricity market design required “the use of two-way contracts for difference (CfDs) for new investments in low carbon generation where public funding is needed” as a means of curbing excessive generator revenues in times of volatile electricity markets.

In 2023, in the Middle East and North Africa, competitive tenders (open or for specific sites) were issued in the UAE, Israel, Algeria, Morocco, Tunisia, Saudi Arabia (3.7 GW) as well as Eritrea,

Voted in May 2024, the European Commission’s Net Zero Industry Act (NZIA) opens the path for support schemes that “promote high sustainability and resilience contribution(s) [...] by either providing additional proportionate financial compensation, or by conditioning the eligibility” (Art 28.1). Non-price criteria can be integrated into competitive tenders and auctions, and the resilience criteria allows governments to impose a diversification of sources of components to be used in a PV system to avoid dependency on a single supplier or suppliers of a single country. This criteria can be used to actively select for local content.

Botswana, Mauritius, Namibia and South Africa (1.8 GW). In some cases, the tenders called for storage also (Panama). In North America the USA ran tenders at state or utility levels. In Latin America, Argentina, Chile, Colombia used tenders – with some significant volumes awarded, for example 5.7 GW in Colombia; in the Caribbean, Jamaica did too. In Asia, many countries continued with their tender schemes including India, Uzbekistan, Sri Lanka, but also the Philippines, Japan and South Korea. Australia, in the Pacific has tenders run by states and utilities.

TRENDS IN TENDERS

As a tool to control and guide towards the type of PV system deemed desirable, governments have adopted new and continuing trends in tenders that include selection criteria or bonuses for local content (Spain), types of land usages or for being installed on buildings (France, Luxembourg), carbon content/footprint (France, South Korea).

Tender specifications can be used to push the development of specific market segments, whether it be in terms of size (between two thresholds, as in South Korea, France, Poland or in the type of PV mounting with separate tenders or designated volumes in tenders for PV on buildings and ground-mounted systems (France, Germany, Switzerland, Moldova). There is a clear trend to

specific tenders or tender volumes for agrivoltaics France, Israel, Luxembourg in the past, and Italy in 2023). Tenders for floating PV included Portugal in 2022, India, Philippines, South Africa in 2023. It can also be used as a driver for innovation, allowing higher remuneration levels for innovative systems not quite ready for the market, as used in France, Germany.

Environmental or local content constraints are introduced to give an advantage to local companies or to favour a better environmental footprint of the products. Reserved volumes or segment bonuses promote specific technologies or impose additional constraints such as local manufacturing to boost the local industry. In several countries, a local content parameter has been discussed and acts as an additional primary or secondary key in the selection criteria (India, Malaysia, South Africa, Türkiye). This type of requirement aims at enabling the development of local solar module manufacturing. Türkiye, for instance, applies a premium for local content, on top of the normal FiT.

Tenders have mostly been used to frame PV development and PV costs on a national level. For regulators, this implies defining a maximum capacity and selecting the cheapest suitable plants to develop. However, they can be further developed and used to guide towards larger, long-term roadmaps on power capacity and segment development. By planning smartly with transmission grid operators, tenders can allow the development of specific capacities for defined technologies, optimize the grid and anticipate the energy transition as a tool to support local industry. They can also be tools to build public acceptance by setting criteria that satisfy public and social acceptability guidelines – such as reserving support to specific types of land usage.

Other elements that have been trialled include technology-neutral tenders. In this case, PV is put in competition with other generation sources. Some countries such as Canada, France, Germany, Spain and Italy have experimented with mixed auctions based on solar and wind in parallel with some technology-specific tenders.

In 2023, in the Middle East and North Africa, competitive tenders (open or for specific sites) were issued in the UAE, Israel, Algeria, Morocco, Tunisia, Saudi Arabia (3.7 GW) as well as Eritrea, Botswana, Mauritius, Namibia and South Africa (1.8 GW). In some cases, the tenders called for storage also (Panama). In North America the USA ran tenders at state or utility levels. In Latin America, Argentina, Chile, Colombia used tenders – with some significant volumes awarded, for example 5.7 GW in Colombia; in the Caribbean, Jamaica did too. In Asia, many countries continued with their tender schemes including India, Uzbekistan, Sri Lanka, but also the Philippines, Japan and South Korea. Australia, in the Pacific has tenders run by states and utilities.

PV MARKET DRIVERS AND SUPPORT SCHEMES / CONTINUED

COMPETITIVE TENDERS AND THE MARKET

Competitive tenders have driven the worldwide market, giving project developers the security of contracts backed by state support mechanisms to invest in increasingly large volumes, and providing governments with a tool to map and control solar energy capacity increases.

Tendered prices in competitive tenders were steadily decreasing up to 2021 when the downward price trend halted due to module price hikes resulting from a combination of COVID and demand impacts. Since bidders must compete with one another, they tend to reduce the bidding price to the minimum possible and shrink their margins. Developers with a large number of projects in previous tenders will have good visibility on price pressure and are most likely to be able to price their projects to winning bids.

The past years have demonstrated how low the bids can go under the constraint of competitive tenders. Low bids associated with FiT or CfD remuneration could be risky- many experts believe such low bids are only possible with extremely low capital costs, low component costs and reduced risk hedging. The shrinking profit margins, especially in super-competitive tenders, could become a threat to the long-term stability of some market actors, hence creating more market concentration. Because of strong competition, the most competitive (lowest) bids are also often

costed on expected module price drops, leaving these projects fragile and potentially unable to remain economically viable when confronted with rising costs as was the case over 2021/2022.

Across all renewable energies, negative prices for tenders with FiP mechanisms have also been seen in recent years; in these cases, market sales or PPA's provide the project remunerations whilst the tenders framework provides other required keys such as grid capacity, security for debt financing....

Conversely, when module costs go up or market electricity prices go up, upper ceilings on tendered prices can be insufficient (either because they are below required remuneration levels or because they are less attractive than market prices). In this last case, for competitive tenders to remain attractive to project developers, they must once again provide benefits that cannot be found on the market – state backing of long-term contracts is the most obvious advantage, pleasing both equity and debt participants, but also landowners.

There have been many changes in the photovoltaics competitiveness landscape over the past two years, from the rising costs in 2020/2021, to the significant drops in module prices in 2023 as electricity markets stabilised and market prices decreased. As a consequence, the dynamics of project profitability have varied, as has the attractiveness of government-run tenders and PPA's.

EU POLICY FRAMEWORK

In December 2018, the revised European Renewable Energy Directive (RED II) set a 32% renewable energy target by 2030, up from a target of 20% in 2020¹. In 2019 the European Green Deal was introduced, an action plan to boost the efficient use of resources by moving to a clean, circular economy and to restore biodiversity and reduce pollution. A pillar of the European Green Deal is a commitment to be climate neutral by 2050. In September 2020, the European Commission proposed raising the 2030 climate targets aiming at a 55% GHG reduction by 2030. The accompanying impact assessment² showed that such an increase in the climate ambition is realistic and economically feasible. The 55% GHG reduction target will require a share of renewable energy of approximately 38,5% according to the impact assessment.

In the European Union, solar energy and photovoltaics in particular were identified as one of the cornerstones of a rapid and more ambitious deployment of renewable energy technologies in order to meet the climate-neutrality objective in 2050 and a significant reduction of the EU's dependence on imported fossil fuels. Although the currently proposed measures include a strong component to diversify the source of fossil fuel imports, away from Russia, a path on how to phase out entirely their use is not yet so clear. The full implementation of the "Fit for 55" proposals would lower the Union's gas consumption by 30%, still requiring over 200 bcm, by 2030.

In 2022, the European Commission published the REPowerEU Communication and the Solar Strategy Communication respectively^{3, 4}.

1. European Commission, REPowerEU Communication, 08.03.2022, COM(2022) 108 final

2. European Commission, Solar Strategy Communication, 18.05.2022, COM(2022) 221 final

3. European Commission, REPowerEU Communication, 08.03.2022, COM(2022) 108 f

4. European Commission, Solar Strategy Communication, 18.05.2022, COM(2022) 221 final



REPowerEU aims to reduce net emissions by at least 55% by 2030 and the Solar Strategy called for an additional photovoltaic capacity of 450 GWp between 2021 and 2030, which would mean a roughly fourfold increase of the nominal capacity to over 720 GWp by 2030.

The European Solar Communication includes a number of building blocks to accelerate the deployment of PV in a timely manner. The following initiatives have been developed as tools to reach the target outcome:

European Solar Rooftops Initiative

Utility scale deployment including multi use of land (e.g. agri-PV, floating-PV, PV on noise barriers, etc.)

Solar value for buildings, districts and cities

Preparing the energy network for the efficient distribution of solar energy

Establishment of a resilient supply chain

Supporting investments regarding EU PV manufacturing (de-risking, funding)

Enacted in June 2024, the Net-Zero Industry Act⁵ aims to create a favourable environment to scale up manufacturing of net-zero industries in the EU. One of the identified strategic net-zero technologies is PV. A simplification of the regulatory framework (permitting) for PV manufacturing and support for skills development are among the actions included in the Act that aim to help increase the EU PV competitiveness.

5. Regulation (EU) 2024/1735 of the European Parliament and of the Council of 13 June 2024 on establishing a framework of measures for strengthening Europe's net-zero technology manufacturing ecosystem and amending Regulation (EU) 2018/1724 (Text with EEA relevance)

In this context, tenders have been under-subscribed (France, South Africa in 2022, Japan, South Korea, Poland, Serbia in 2023) – for example, when market conditions were more attractive – under-awarded when tendered rates were considered uncompetitive i.e. close to or over ceiling rates (France in 2020 and 2021 for some segments, Spain in 2022) or even over-subscribed where the security of investment and low modules costs made them more attractive (in 2023: Argentina, France, Germany, the Luxembourg/Finland cross border tender, the Luxembourg agrivoltaic tender, UAE).

In 2023 the lowest bids once again decreased, with Japan reaching bids under 8 Yen/kWh (55 USD/MWh) for the first time. The most competitive bids in 2023 were in the UAE at 16.2 USD/MWh (but ranging up to about double in India and more than triple in Europe), compared to 26 USD/MWh in 2022 (India, South Africa, Türkiye) and 60 USD/MWh (utility) to 90 USD/MWh (on buildings) in Western Europe (Germany, France).

Within this context, developers are comfortable looking to corporate clients to sell their electricity (corporate PPA's) (Germany, France, Spain, USA), or even directly to electricity markets (merchant PV) (Australia, Croatia, Germany, Romania, Spain).

TRADE OF GREEN CERTIFICATES AND SIMILAR SCHEMES

Green certificates and similar schemes based on Renewable Portfolio Standard (RPS) are only used in a few markets, explained by the greater complexity of this type of scheme. Green certificates with a set value can be issued within the framework of government or state support mechanisms (Belgium (residential systems), Australia, South Korea, Norway, UK). They can also be traded at market values between generators and utilities as part of RPS obligations (USA – predominantly but not only utility-scale systems). The regulatory approach commonly referred to as RPS aims at promoting the development of renewable energy sources by imposing a quota of RE sources. The authorities define a share of electricity to be produced by renewable sources that all utilities must adopt, either by producing themselves or by buying specific certificates on the market. When available, these certificates are sometimes called “green certificates” and allow renewable electricity producers to get a variable remuneration for their electricity, based on the market price of these certificates.

Green certificates can also be bought to satisfy voluntary social responsibility goals (in this case, most often Guarantees of Origin (GO) are attached to electricity sold to consumers with a surcharge for the guarantee). Their value can vary depending on the local markets and how active they are – but the 2023 trend was to

PV MARKET DRIVERS AND SUPPORT SCHEMES / CONTINUED

increased value. Some important markets include North America, UK. Depending on the local markets, trading GO's may or may not be cumulative with remuneration from government-financed support mechanisms.

PV DEVELOPMENT WITHOUT FINANCIAL INCENTIVES

An increasing volume of the market became independent of support schemes: this implies installations not financially supported and developed outside of tenders or similar schemes. This is a sign of the PV market becoming highly competitive. The increase in energy costs in 2021 and 2022, and specifically electricity prices, enhanced PV competitiveness in numerous countries. Whilst electricity prices dropped over 2023, so too did module prices; in parallel, consumers exposed to the volatile electricity market over the past years have increasingly turned to securing long-term supply contracts or generating their own electricity, financing PV outside of all support mechanisms.

POWER PURCHASE AGREEMENTS

Power Purchase Agreements (PPAs) are long-term private contracts between a PV producer and one or several off-takers: consumers (Corporate PPA) or electricity resellers (PPA). While FiT (or CfD, in the case of virtual PPA's) are paid in general by official bodies or utilities, commercial PPAs are contracts between the PV plant owner and an off-taker for the electricity produced, over a defined period. Such contracts guarantee a certain level of revenue and are increasingly popular for unsubsidized PV because the cost per kWh is negotiated between the parties. If built on a FiT model, in times of high electricity prices (2021, 2022), the PV producer has an advantage if the contract was negotiated in times of lower prices. PPAs imply sourcing of solar electricity without necessarily being physically connected to the power plant - a solution favoured more and more by large companies willing to decrease their GHG emissions through corporate PPA's. In the case of virtual PPA's operating on a CfD, the off-taker pays the generator (or receives from the generator) the difference between the agreed-on price and the price sold on the electricity market sales; in this case, the off-taker carries the risk burden for the generator. Initially deployed in the wind industry, solar PPA's have outstripped wind PPA's in volume in 2023. PPAs have continued to grow in 2023 after strong growth in 2022 and 2021. Major information technology companies, online retailers and industry are significant clients for virtual CPPA's whilst mining companies are significant clients for PPA's from systems directly connected to consumption sites. The European Union incites member states to remove administrative barriers to long-term PPA and to facilitate their adoption. One of the principal barriers to developing PPAs is the multitude of risks, from low production to off-takers insolvency, and there is a small but growing market in risk hedging; France launched a credit

guarantee scheme to support smaller actors for risk mitigation joining Spain and Norway who also have mechanisms in place. Whilst PPA contracts are mostly medium to long-term (10 to 25 years), the high electricity market prices in 2022 and early 2023 saw a surge in short-term (2 to 3 year) contracts as producers look to the highest returns.

In Europe, there was strong continued growth in the announcement of photovoltaic PPAs with both corporate and utility PPAs increasing. In 2023, Spain and Germany lead the PPA market that surpassed 10 GW in Europe, with active markets also in Denmark, Finland, France, Italy, the Netherlands, Poland, Sweden, and the UK. Solar PPA volumes contracted slightly in the USA, although this still estimated to represent nearly 50% of global PPA volumes. In South America, an emerging market for PPA's exists (over 500MW in Brazil and 244 MW in Chile). The Asia-Pacific region had PPA growth in South Korea (x 10 compared to 2022), Australia (where resource mining and heavy industry account for roughly 1/3 of Corporate PPAs in 2023) and Japan (where electricity charge increased and companies choose to select renewable energies) but also Bangladesh, Philippines, Chinese Taipei and Thailand. Countries that have been working towards structural changes to electricity markets to allow PPAs or corporate PPA's (Vietnam, Malaysia) were good to go in 2023. In the MENA region,

Whilst many PPA developers and buyers are industry specific, there is also a wide range of global energy players present, as discussed in Chapter 4.

MERCHANT PV

Merchant-based PV plants are expected to play a growing role in the development of the PV market. They are PV plants where the business model relies on sales on electricity markets. The design of the electricity market plays an important role in the emergence of this type of business model as the market should provide both short-term and long-term incentives. Whilst several countries experimented with merchant PV as long ago as early 2010 (unsuccessfully in Chile, for example), more recently competitive LCOE and the higher electricity prices on the European market in 2022/early 2023 encouraged project developers to investigate this option more seriously – it is now an option that has been used in Croatia, Italy, Germany, Norway, Portugal, Romania, Sweden and Spain, (where it is expected to become as common as PPA's in the near future) , as well as in the Philippines, Australia (a 2022 study indicated 18% of its then nearly 20 GW capacity exposed to spot markets) in Asia Pacific with projects underway or in the development phase in Botswana, Namibia, in Africa. In the USA as PPA contracts become shorter and shorter (under 10 years), merchant sales are becoming an integral part of the long-term economic viability of projects.



NON-INCENTIVIZED SELF-CONSUMPTION

Self-consumption ceases to need incentives when the revenue from the savings on electricity bills (the self-consumed part) and the revenue from the sale of excess PV electricity covers the long-term cost of installing, financing and operating the PV system. As retail and wholesale electricity prices rise and PV LCOE reduces, self-consumption has become an obvious choice in many markets – particularly in European and Australian markets or markets where electricity prices were disrupted due to the Ukraine conflict.

Where self-consumption becomes the norm, grid managers can become wary of transfers in the burden of grid costs between consumer/producer categories, or even revenue loss where revenue is proportional to consumption. In this context, fixed fees or self-consumption taxes and penalties can reduce the attractiveness of self-consumption and impact uptake. For example, concerns over such cost-shifting from PV to non-PV customers in the USA were cited as one of the main drivers of changes to California's net energy metering revision in 2022, where grid participation charges and minimum bills were initially considered. This in turn led to a rush of installation in California, as consumers rushed to remain eligible to the pre-reform mechanisms.

As grid capacity becomes saturated, triggering curtailment and balancing cost increases, consumers will increasingly look to alternatives, from storage to energy sharing through collective self-consumption or energy communities to improve the economic equation.

INNOVATING FINANCIAL SOLUTION SUPPORT

An increasing number of investment solutions have emerged for the financing of solar installations, these are even more relevant in the case of unsubsidized PV. The high upfront capital requirements are pushing different business models to develop, especially in the USA, and to a certain extent in some European countries. PV-as-a-service contributes significantly to the USA's residential market for instance, with the idea that PV could be sold as a service contract, not implying the ownership or the financing of the installation. These business models could deeply transform the PV sector in the coming years, with their ability to include PV in long-term contracts, reducing the uncertainty for the contractor. Such business models represent already more than 50% of the residential market in the USA, and some utilities in Germany, Austria, Sweden and Switzerland are starting to propose them. The USA remains innovative by the existence of pure players proposing PV as their main product, solving many barriers to financing and effective operations, as well as reducing the uncertainty in the long-term for the prosumer. It is possible that such services will further develop.

Similarly, pay-as-you-go financial models have been very successful in the deployment of Solar Home Systems (SHS) and solar kits in African countries in the past years and are expected to further drive the development of PV in the residential and off-grid segments. Pay-as-you-go models are directly inspired by prepaid mobile payment schemes; the users pay a monthly fee according to their needs and own the solar kit when enough credits have been paid.

PROSUMER AND ENERGY COMMUNITY POLICIES

SELF-CONSUMPTION IN REGULATORY ENVIRONMENTS

Prosumer (self-consumption) regulations are increasingly being implemented in different countries with the double goal of empowering consumers to play an important role in the energy transition and reduce the cost of support mechanisms for the development of renewable energies. Measures in favour of distributed generation are stimulating greater use of renewable sources with further positive effects such as a stronger penetration of electricity in final consumption, the reduction of transmission and distribution costs and new investments in integrated energy management projects (electricity, heat, efficiency, storage, etc.).

Self-consumption is allowed one way or another in many countries but the regulations in place differ significantly with both structured and ad hoc legal frameworks, although the physical principles of self-consumption are always the same: the electricity that is produced by the PV system and locally consumed reduces the quantity of electricity on the consumer's bill. However, this reduction is not implemented in the same way in all countries.

Mechanisms to promote or regulate self-consumption tend to address one or more of the following:

1. Initial grid connection and/or annual grid access fees
2. The right to inject excess and the remuneration of excess electricity.

Grid connection fees can be reduced or waived to promote self-consumption (especially when consumers are already equipped with electronic meters that require no direct intervention) (France, Finland, Netherlands) or on the contrary adapted to cover self-consumption. As in most countries grid access fees are calculated with at least part being proportional to consumption, a number of different approaches have been taken (or are under discussion) to diminish the impact on grid revenue for distribution service

PROSUMER AND ENERGY COMMUNITY POLICIES/ CONTINUED

operators. Some adaptations adopted or discussed include: modified grid tariff structures (increasing the fixed part and reducing the variable part linked to consumption (Australia, France, Switzerland); specific grid taxes to compensate for saved grid costs due to net-metering/billing policies (Israel, Spain, Belgium); minimum bills to ensure a minimum level of remuneration from the user (USA). It should be noted that states and utilities in Australia and the USA operate distinctly, and a variety of mechanisms have been used.

The right to inject excess electricity has required structural changes to many regulatory frameworks around the world as it intersects with generation and electricity sales on the one hand and balancing requirements on the other. In high penetration grids, or under-dimensioned grids, automatic rights to inject can also have serious consequences in terms of grid stability management when consumption is low and injections are high (Australia, Austria, some states in the USA, Vietnam). In these countries, limits on injection (injection capacity or time of injection) can be put in place; increasingly, distribution service operator-led dynamic curtailment of injection limits for distributed generation and prosumers is being investigated to ensure both grid stability and smooth development of PV (Australia, USA, Japan, Spain).

The excess electricity can be credited to the consumer (net metering) on a specific timestep (monthly, quarterly, annually) with unused credits either carried over or lost; it can be sold (net billing on short timesteps, from 5 minutes up to several hours or a day) to the utility, to support mechanisms buyers or on the market; it can also earn green certificates that can be traded. See above for more information.

Finally, third-party ownership (often synonymous with long-term leasing) allows the development of self-consumption even when consumers do not have access to the capital necessary for up-front costs – whilst it is common in some countries (USA), in others the complex regulatory environment means it is either not possible or requires changes to legislation (Denmark, France, Spain).

COLLECTIVE SELF-CONSUMPTION, ENERGY COMMUNITIES AND COMMUNITY SOLAR

Collective self-consumption, also called delocalized or virtual self-consumption, energy communities or community solar, is when the electricity from one or more generators is allocated to one or more consumers, either reducing or offsetting energy consumed from other suppliers through the grid. The energy flows can be physical (for example when a PV system injects to internal grids in multiple-tenancy buildings) or virtual (when generation is allocated via virtual metering and billing, using a predefined split key). The price of the electricity can be based on a negotiated per kWh fee or accessed after a share in the capital investment.

It began as a single system supplying on-site consumers with self-consumption; from there, collective self-consumption developed with the aim of widening the perimeter of self-consumption. This allows one or several PV producers (even utility-scale plants) to feed one or more consumers at a reasonable distance so that the use of the public grid is minimized. The perimeter, often initially limited to a building, has expanded in different countries to a geographical distance (10 km, 20km radius...or across a district, region or the whole country) or set on a transformer or substation level. In this sense, some schemes compensate for real energy flows, while others are compensating for financial flows. While details may vary, the basics are similar. This widening of the perimeter to the district or regional level to include several consumers and generators has advantages for participants that include increased self-consumption ratios and more equitable access to roofs and land for self-consumption purposes (Mexico, Brazil, France, Lithuania, USA).

The economic viability of collective self-consumption projects is built not only on retail and wholesale electricity prices and generation prices but also on the contributions participants must pay to grid access and the level of taxes on consumption and generation.

There has been strong interest in particular in Italy, France, Spain, the USA and to a lesser extent Portugal, Austria, Canada, Norway Sweden, Switzerland, Germany.



In Italy, there are incentives for systems up to 1MW in groups behind the same substation. In Sweden, it had been allowed through microgrids in 2021. In France, virtual self-consumption operations can cover a perimeter of up to 20 km (rules apply) and there are simplified administrative and legal procedures for social housing operators running operations on multi-tenancy buildings; generators can access the FIT for excess electricity not immediately self-consumed within an operation's 30-minute compensation time step. In Germany, building owners can produce and sell electricity to their tenants which makes the investment more attractive. The UK has also implemented a favourable framework for collective prosumers. In Spain public policies and subsidies promote energy communities. Within the USA, the Inflation Reduction Act included significant tax credit bonuses for collective self-consumption projects that met certain size, location, and equitable distribution of benefits requirements; nearly half of the states have passed legislation enabling virtual collective self-consumption (referred to within the USA as "community solar"), with many including requirements for participation of low-income households. Other countries have some definitions, but these are not yet fully implemented. In Austria, electricity traded in energy communities benefits from tax exemptions. In Switzerland collective self-consumption is allowed by most DSOs, but consumers have to be contiguous and not use the public grid. In Norway, since 2023 consumers on the same property can share and self-consume electricity produced from a PV system of up to 1 MW, and discussion is underway to allow virtual self-consumption in industrial areas. In the USA and Australia community and edge-of-grid rural microgrids are emerging to reduce the cost of electricity consumption and provide local resilience through storage and backup power.

Network pricing regulations differ, and, in some countries, exemptions are used to encourage projects whilst in others full network charges must be paid even for locally transmitted electricity, which acts as a barrier to collective self-consumption or virtual net-metering.

These schemes create complex questions, especially regarding the use of the grid, the legal aspects and technical complexity related to compensating electricity between several meters and the innovative aspect of the scheme.

The opportunities opened up by such concepts are wide-ranging and could tend to increase consumer participation in market mechanisms and PPA participation. As an extension, consumption metering points could become virtual too - innovative schemes could allow charging an electric vehicle at the office with PV electricity produced at home or sharing the PV electricity in all public buildings in a small town between them depending on the

consumption. Options are numerous and imply fair remuneration of the grid to be competitive for all. Using PV electricity in a decentralized location implies the use of the public grid, distribution or even transmission and would require putting a fair price on such use.

PROSUMER AND ENERGY COMMUNITY POLICIES / CONTINUED

in the EU context

While self-consumption is allowed in most European countries, Europe has decided to go a step further with the comprehensive update of its energy policy, the “Clean Energy Package”. The European Union introduced new provisions on the energy market design and frameworks for new energy initiatives. Specifically, the actual recast of the renewable energy directive (REDII) and the electricity market directive (EMDII) provide basic definitions and requirements for the activities of individual and collective self-consumption. The European Union introduced the concept of Renewable Energy Communities (REC) and Citizen Energy Communities (CEC). REC should allow citizens to sell renewable energy production to their neighbours, while some crucial components are the definition of the perimeter and the tariffication for grid use. Those key components are defined in the national implementation in the member states that are being deployed across Europe. This concept of energy communities is expanding the existing PV market segments and allow cost reductions for consumers not able to invest in solar installation themselves. In Europe, the Federation of energy communities REScoop has over 2000 member organizations representing more than 1.5 million citizens, testifying to the strong development.

in the USA

A different definition of an energy community is emerging in the USA as a result of the passage of the Inflation Reduction Act. It is now used to refer to communities that have been historically adversely impacted or are at risk of being adversely impacted in the future by the energy transition. The use of the term Community Solar in the USA is closer to the definition of collective self-consumption than the EU REC. Solar communities are growing under the impetus of IRA subsidies and state-led legislation – with often sharp crossovers joining community solar to energy communities; for example, Maryland, Minnesota and New Jersey have similar programmes that require community solar facilities to deliver a minimum percentage of their output to low- and moderate- income subscribers. By the end of 2023, community solar had an estimated capacity of 7.3 GW AC, with 75% of this power in just four states (Florida, New York, Massachusetts and Minnesota).

ENERGY TRANSITION LEVER POLICIES

SUSTAINABLE BUILDING REQUIREMENTS

The building sector has a major role to play in PV development and sustainable building regulations drive PV's deployment in countries where the competitiveness of PV is close. These regulations include requirements for new building developments (residential and commercial) but also, in some cases, on properties for sale. PV may be included in a suite of options for reducing the energy footprint of the building or specifically mandated as an inclusion in the building development.

The publication of the European Commission's Solar Strategy in 2022 is part of the REPowerEU package. It presents four initiatives to overcome the remaining short-term challenges and the first of them is promoting quick and massive PV deployment via the European Solar Rooftops Initiative. Member states are incorporating this initiative in different ways into their national regulations.

In Austria, the federal support schemes have a bonus for building

integrated PV whilst many countries have regulations or incentives for building a PV system, with mandatory solar in Vienna and Styria for commercial and apartment buildings. France has mandatory solar or living roofs for commercial and industrial buildings or covered car parks. In Germany, in Berlin, solar will be mandatory on many new buildings by 2023 whilst in the Netherlands, buildings must aim to be nearly energy neutral since 2021, pushing solar. In Switzerland, mandatory solar is being rolled out across different cities for new and/or renovated buildings. In Belgium, Flanders introduced measures in 2014 with PV now on more than 85% of new buildings. In Denmark, the national building code has integrated PV to reduce the energy footprint, whilst in Italy, capital subsidies have been introduced to promote PV on public buildings in some regions. In Japan, Tokyo will be requiring solar on most residential buildings, and other smaller buildings from 2025. In South Korea, the NRE Mandatory Use for Public Buildings Programme imposes on new public institution buildings with floor areas exceeding 1000 square meters to source more than 10% of their energy consumption from new and renewable sources. In the USA, California has had mandatory solar for certain new residential buildings since 2020



and is set to extend this to non-residential and high-rise multifamily buildings in 2023 and expand the mandate to require a solar plus storage system, rather than just a solar system. In India, some states have mandatory solar supply policies for new buildings.

Beyond overall building performance regulations, rising questions around the public acceptability of utility-scale PV is pushing a number of governments to actively encourage or mandate solar in very particular segments – from Japan’s program supporting research, guidelines and demonstrators for wall-integrated PV (including semi-transparent and perovskite cells) to France’s mandatory solar on impermeable car parks. The goal of occupying low-conflict surfaces is pushing solar too.

ELECTRIC MOBILITY

The electrification of transport is accelerating in many countries and whilst the link between PV development and EVs is not yet fully understood, the growth of self-consumption policies and grid congestion limiting injections are factors to be considered. Charging EVs during peak load implies rethinking power generation, grid management and smart metering, and concepts such as virtual self-consumption could rapidly provide a framework for EV’s as mobile storage for excess PV generation. With 14.2 million EVs sold in 2023 (+35% on 2022), electric vehicles sales are still accelerating in parallel, albeit at a different rhythm, to PV. Growth was particularly dynamic in some IEA PVPS countries including, Australia, Canada, China, Spain, Switzerland and Türkiye, Thailand, Malaysia. Non-IEA PVPS countries with strong growth markets include India and Brazil. In Europe, EU directives aim for 100% zero-emission vehicle sales by 2035 (2030 in Austria, the Netherlands, Denmark) and national and local-level actions exist to support EV deployment, both with incentives for those buying EV’s or investing in charging stations, but also in terms of barriers for traditional internal combustion vehicles (France, Norway, Switzerland). In Austria, a federal program supports the purchase of electric vehicles for private – on the condition that it is supplied by 100% renewable energy. In Germany a support programme for solar for charging EV’s launched in 2023 closed after its budget was depleted in just one day. Outside of Europe, many countries with dynamic PV markets also have strong financial incentives for electric passenger vehicles and charging stations (Japan, some states in Australia, India). In 2022, the USA implemented significant national incentives for the purchase of EVs via the Inflation Reduction Act and began the buildout of a national EV-charging network via the Infrastructure, Investments, and Jobs Act of 2021. In South Korea public entities may only purchase eco-friendly vehicles (electric or hydrogen), whilst local and national government support charging stations roll out and energy storage businesses with installed solar can now sell power directly to electric vehicle users.

Countries with higher penetration rates or good uptake, such as Norway, Sweden supported the roll out of electric vehicles for many years but have already stopped incentives. China no longer offered rebates in 2023 but maintained tax exemptions. Germany stopped its subsidies in 2023. In Australia, one state introduced a tax specifically for EV users to offset taxes no longer being collected through fuel levies in 2021; this tax was deemed unconstitutional in 2023 and will be reimbursed to EV owners.

HYDROGEN PRODUCTION

Solar fuels, storage and other hydrogen-based applications will require massive PV, wind and other RES development if hydrogen is to be “green”. Distributed Hydrogen production could be driven by distributed PV as well, pushing for higher demand for distributed PV-H₂ production. This still remains to be materialized in the future, however strong interest after the Ukraine conflict-hit gas and electricity prices in 2022 has been supported by strong government investment programs. In the USA, the Inflation Reduction Act financed a hydrogen hub program and supplies tax credits for hydrogen produced from renewable energies including solar. The European Commission expects green hydrogen to play a pivotal role in the decarbonisation of sectors where electrification might be less feasible and to bridge some of the gaps for seasonal variations of electricity generation (ie from solar). Funding programs promote research and a pilot project to reach the ambitious plan of between 15 GW and 40 GW of electrolyzers in Europe by 2030. As member countries were required to submit updated National Energy and Climate Plans plans in 2023, nine member countries included funding and tax incentives for hydrogen projects; Spain tripled its 2030 targets for renewable-based hydrogen, Portugal doubled theirs. In China the biggest solar + hydrogen project was commissioned in 2023 and should produce 20 000 tons/year of hydrogen. In India competitive tenders for electrolyzers for green hydrogen were launched in 2023, with a called capacity of 1.5GW of electrolyser capacity in the first round.

Large-scale renewable + hydrogen production projects finalised financing around the world in 2023: several each in Australia, Germany and France – whilst not exclusively solar-powered, cheap solar is an important part of the economic equation for these projects along with wind power.

ELECTRICITY STORAGE

The cost of storage is pursuing its steep decline after a short halt in 2022 - and storage is becoming more attractive in a growing number of markets. Due to the cost decline of storage, solar power plants with onsite storage are increasingly attractive for developers and residential customers as the combination with storage allows them

ENERGY TRANSITION LEVER POLICIES / CONTINUED

to smooth the power output, to deliver ancillary services or to reduce connection costs if peak injection is reduced.

In China, capacity grew significantly in 2023 as in many provinces' storage is mandatory for new utility-scale solar, whilst commercial installation also grew due to changing time of use consumption tariffs and the availability of subsidies. Incentives for battery storage exist in a wide range of countries, including Austria, Japan and Romania whilst plans are being made for the near future (Italy, Hungary, Canada). Spain has launched a call for renewables + storage projects, subsidising storage for self-consumption projects; there are reserved volumes for projects on the Canary and Balearic Islands. In the USA tax credits are available as part of the Inflation Reduction Act, whilst multiple states have energy storage procurement goals or mandates, and in California solar + storage is now mandatory for many types of new commercial or multifamily dwellings. In Australia, households have access to rebates in some states, whilst in others time of use and power injection caps have given battery storage short payback times. Unattached to solar but enabling high PV penetration rates in the grid, Federal government support for Big Batteries continues with a revenue underwriting scheme (similar to CfD) and some states have grants and funds for market positioning and capacity building.

Storage is a key element of a carbon neutral energy system relying on RES electricity; therefore, the European Commission actively supports energy storage through research and innovation funds. Some consider that storage development for PV electricity will be massively realized through electric vehicles connected to the grid during a large part of the day and therefore, will be able to store and deliver energy to consumers at a larger scale than simple batteries. These vehicle-to-grid or V2G concepts are being explored and tested in several countries, with the Netherlands, Switzerland and Japan as front-runners, with at least one V2G project selling on the electricity market.

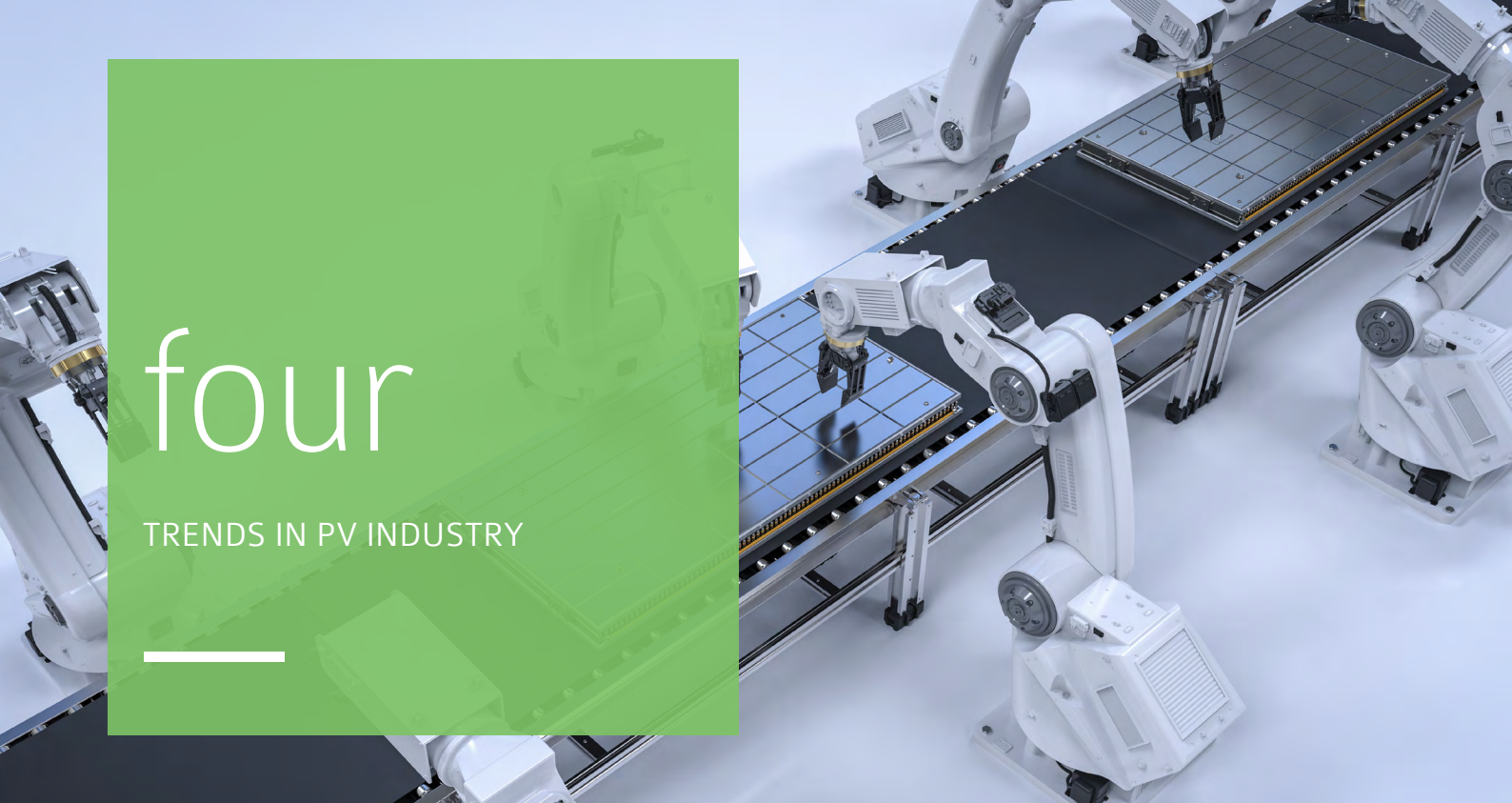
INDUSTRIAL AND MANUFACTURING POLICIES

COVID-led supply chain disruptions and increasingly ambitious climate change actions saw many initiatives favouring local manufacturing at various steps of the PV value chain in 2021 and 2022, with governments around the world supporting local manufacturing through policies, subsidies and regulations – from tax credits to direct subsidies to reduced administrative barriers for local content. However, the drastic reduction in module prices in mid-2023 as new capacity came online in China, well above the international market's ability to absorb, has made the future of many projects uncertain. Some of the most significant acts of the past decade were launched

in 2022, including the Inflation Reduction Act (IRA) in the USA and the preparation of a series of EU-level policies in Europe. The IRA was promulgated in August 2022 and the generous tax incentives for local manufacturing, critical mineral refining, and recycling have triggered many announcements for new manufacturing capacity in the USA. In Europe a series of policies planned in 2022 but published in 2023 are intended as landmark acts in the EU Solar Energy Strategy (within the REPowerEU plan) – the Green Deal Industrial Plan (published in February 2023), the Net-Zero Industry Act (NZIA) and the Critical Raw Materials Act (both published in March 2023), and the Net Zero Industry Academies were enacted, stimulating and encouraging local manufacturing plans. Future changes to the Carbon Border Adjustment Mechanism may also have positive impacts in Europe. Amendments to the Temporary Crisis Framework at the end of 2022 lift limitations for the EU Member States to grant direct CAPEX support to new net-zero industry projects, with solar manufacturing being entirely eligible. Co-funded by the EU and launched in late 2022, the European Solar PV Industry Alliance aims to facilitate innovation-driven expansion of a resilient industrial solar value chain in the EU, particularly in the PV manufacturing sector. These EU level actions have been complemented by national measures planned over 2022 and 2023 in the Netherlands, Spain Italy, which came into action in 2024. In India, state financing for local manufacturing is available through the Production Linked Incentive Scheme that was continued in 2022, with three winners in Round 1 – and more than ten announced in further rounds in 2023, a first step to building dozens of GW of domestic manufacturing capacity capable of supplying a large proportion of the local, growing PV market. In Türkiye a new YEKA (Renewable Energy Resource Area) call for tenders is expected but not published in 2022 – the YEKA calls give priority to areas and RES projects that can be supplied with domestic manufacturing and have proved successful so far.

In China industry and government have acted to maintain their position as the pre-eminent manufacturer of PV, and with massive private investments over 2022 they have in fact triggered overcapacities in large-scale manufacturing, bringing down module prices to below cost in 2023 and creating a natural barrier to entry for new manufacturers in other countries.

Whilst there is a clear push to develop support for local manufacturing across the world, there is not always a clear understanding of the industry dynamics and the complexities of PV manufacturing, which will lead to fewer real projects than what some governments would like to see. In addition, as the proportion of global supply of specific raw materials (glass for example) devoted to PV increases, local manufacturing will require access to global value chains and the role of already existing global actors should not be neglected.



four

TRENDS IN PV INDUSTRY

This chapter provides a brief overview of the upstream and downstream sectors of the PV industry intending to provide highlights during 2023 and the first half of 2024. The first part provides manufacturing activities of the upstream sector of the PV industry from feedstocks (metallurgical grade silicon, (MG-Si), polysilicon, ingots, blocks/bricks and wafers) to PV cells and modules described in Figure 4.1. The second part provides activities of the balance-of-system (BOS) sector that include components (inverters, mounting structures, charge regulators, storage batteries, appliances, etc.), and project development and operation and maintenance (O&M).

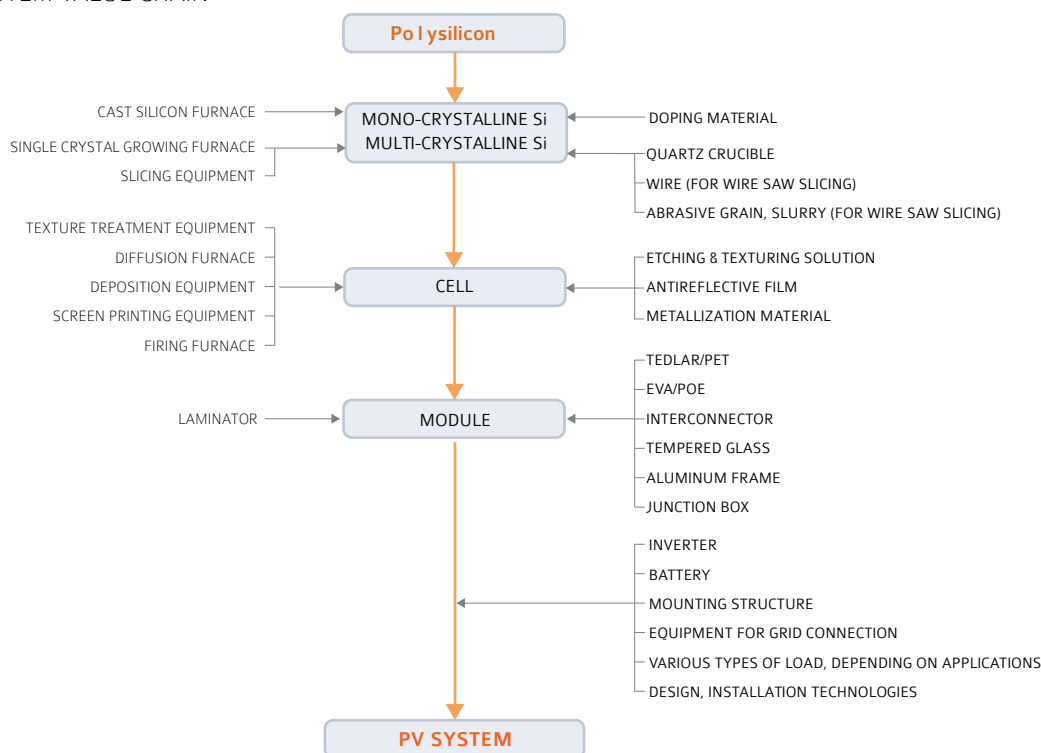
In 2023, the gaps between demand and manufacturing capacity affected lowering price trends along the value chain of PV modules. The speed of manufacturing capacity expansion surpassed the market growth and the increase of module inventories also affected price levels. The price of polysilicon dropped from 22.13 USD/kg at the end of 2022 to the 7 USD/kg range by the end of June 2023, and it remained below 10 USD/kg thereafter. PV module prices were 11-12 US cents/W as of the end of December 2023. While concerns about rising project costs in 2022 have been resolved, PV module suppliers have been facing lower margins and cost pressure.

Throughout the year, projects for increases in manufacturing outside of China have been reported in the USA, India and Europe. Due to the Inflation Reduction Act (IRA), it is reported that the USA production capacity of PV solar modules will exceed 100 GW/year in 2026. Projects for manufacturing inverters and trackers are also planned there. In India, due to the PLI, the solar module production capacity is expected to reach 60 GW/year by the end of 2025. Despite these projects, China remained the world's largest producer along the PV supply chain and further increases of

manufacturing capacity in China were reported in 2023 whilst the Chinese production share increased in all of the value chain.

In 2023, production capacity increased to 1 103 GW/year from 717 GW/year in 2022. The production capacity figures include the capacities of aging and idle facilities that are not competitive anymore: consequently, the real, effective, production capacity is assumed to be on the level of about 700 GW/year to 800 GW/year in 2023. The speed of capacity enhancement is faster than the global PV market growth so the gap between demand and supply was wider in 2023 and PV module prices remained lower. If the gap is not bridged and inventories are not cleared, current price levels are expected to continue.

FIGURE 4.1: PV SYSTEM VALUE CHAIN



SOURCE IEA PVPS & OTHERS

THE UPSTREAM PV SECTOR

SILICON METAL PRODUCTION

Wafer-based c-Si technology remains dominant for producing PV cells. In that respect, this chapter focuses on the wafer-based production process. Silicon metal (Metallurgical grade Si) is used as a raw material for polysilicon, which is used to make ingots and wafers then processed to solar cells. Sustainability of the industry in terms of resources availability and fair business practices, and silicon metal production became focuses of the PV industry. In 2023, the world manufacturing capacity of silicon metal reached 8.54 million tons/year. Amongst the manufacturing countries, China has the largest capacity at 7 million tons/year (82%). As well as in the other steps of crystalline silicon solar cell manufacturing, China has the dominant position in this sector. In 2023, China produced 3.76 million tons of silicon metal, and domestic consumption was 3.19 million tons, of which 53% (1.825 million tons) was used to produce polysilicon. China foresees high growth in demand for silicon metal. Production in 2024 is expected to increase 20% year on year to 4.7 million tons in 2024. In China, Xinjiang has the largest

share of production (48%) in 2023 followed by Yunnan (17%) and Sichuan (16%). Some major Chinese PV companies are planning to invest in production. For example, Tongwei group has a plan to establish two plants (360 000 ton/year and 400 000 ton/year) in Inner Mongolia. Trina Solar invested to build vertically integrated factories in Qinghai Province, including silicon metal (100 000 tons/year). Other companies such as Juhua Group are also reported as having new investment plans. Outside of China, new investment plans are reported reflecting the needs to diversify the PV supply chain. In India, Adani New Industries (ANIL) has raised \$39.4 billion to set up vertically integrated factories for silicon metal, polysilicon, ingots, wafers and solar cell and PV modules (with a production capacity of 10 GW/year). In Australia, Solquartz is planning to produce MG-Si and polysilicon in the same area.

POLYSILICON PRODUCTION

The global polysilicon production (including semiconductor grade polysilicon) in 2023 was about 1 608 000 tons, a 61% increase from the previous year. Production of polysilicon for semiconductors was about 38 800 tons. This means that more than 98% of polysilicon production was used for PV applications. Global polysilicon production capacity (including production capacity for semiconductors) increased by 83% from 1 354 700 tons/year in 2022 to 2 487 400 tons/year in 2023.